COMSATS University Islamabad, Sahiwal Campus

HOME2U

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Bachelor of Science in Computer Science (2021-2025)

The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.

COMSATS University Islamabad, Sahiwal Campus

HOME2U

A project presented to

COMSATS Institute of Information Technology,

Sahiwal Campus

In partial fulfillment of the requirement

for the degree of

Bachelor of Science in Computer Science (2021-2025)

By

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DECLARATION

We hereby declare that this software, neither whole nor as a part has been copied out from any source. It is further declared that we have developed this software and accompanied report entirely on the basis of our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other. We will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

Ayesha Abdul Quddoos Khan | Zainab Zia

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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) “Home2U” was developed by **Ayesha Abdul Quddoos Khan (CIIT/FA21-BCS-187/SWL)**, **Zainab Zia (CIIT/FA21-BCS-200/SWL)** and **Burhan Ahmad Khan (CIIT/FA21-BCS-216/SWL)** under the supervision of “**Ms. Anum Khan**” and co supervisor “**Ms. Raheela Shehzadi**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in Computer Sciences.

Supervisor

External Examiner

Head of Department

(Department of Computer Science)

Executive Summary

In today's real estate market, buyers face significant challenges including unverified property listings, reliance on static images, and inefficient communication channels. Traditional platforms allow both buyers and sellers to post listings, which often leads to outdated information, fraudulent posts, and a lack of standardization. Moreover, the inability to properly visualize properties remotely results in unnecessary physical visits and wasted time.

To address these problems, HOME2U was developed. It is a website that provides solutions to many of these issues. The platform features administrator-controlled listings to ensure all property information is accurate and up-to-date. A key innovation is the integration of price prediction model and 360-degree virtual tours, allowing buyers to thoroughly explore properties from anywhere and estimate price. Advanced search filters enable users to quickly find properties matching their specific criteria, while secure communication channels facilitate direct contact with the administrator.

HOME2U is built using modern web technologies, including React.js for the frontend, Node.js for backend operations, AWS for secure image storage, and MySQL for database management. The platform is optimized exclusively for desktop use, delivering a consistent and user-friendly experience.

Unique aspects of this system include:

- Exclusive administrator control over property listings to ensure data authenticity
- Integrated AI-powered model that predicts property prices based on area, location, and room details
- Immersive 360-degree virtual property tours not commonly found on competing platforms
- Advanced filtering options that go beyond basic location and price parameters
- Encrypted communication channels between buyers and administrators

The website represents a significant step forward in real estate technology, combining verification mechanisms with innovative visualization tools to create a more transparent, efficient property buying process. Future enhancements could expand the platform's

capabilities to include mobile access, artificial intelligence recommendations, and blockchain-based transaction security.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

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Ayesha Abdul Quddoos Khan

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Abbreviations

SRS	Software Requirement Specification
PC	Personal Computer
SDD	Software Design Description
AWS	Amazon Web Services
OWASP	Open Web Application Security Project
SSL/TLS	Secure Sockets Layer/Transport Layer Security
CRUD	Create, Read, Update, Delete
UI/UX	User Interface/User Experience

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Introduction

1 Introduction

HOME2U is a modern real estate platform designed primarily for property exploration, offering users the ability to browse and evaluate listings with ease and efficiency. While future enhancements will include comprehensive buying and selling functionalities, the current focus is on providing a seamless experience for homebuyers, property investors, and real estate agents alike. Built using cutting-edge technologies such as React.js, Node.js, AWS, and MySQL, HOME2U delivers a desktop-optimized interface with user-friendly navigation that ensures even those with minimal tech experience can effectively utilize the platform without feeling overwhelmed. The design prioritizes accessibility, allowing users to navigate through various listings effortlessly, whether they are searching for a cozy apartment in the heart of the city, a spacious family home in a suburban neighborhood, or a sprawling estate in the tranquil countryside.

The platform features AI-powered price prediction, which analyzes a myriad of factors, including recent market trends, neighborhood characteristics, property features, and historical data, to offer users accurate estimates of property values. This innovative feature empowers users to make more informed decisions, enhancing transparency in the valuation process and significantly reducing the likelihood of overpaying for a property or underselling their own. By incorporating these predictive analytics, HOME2U not only aids in the financial aspects of real estate transactions but also builds a foundation of trust between users and the platform, as they can rely on datadriven insights rather than mere speculation.

In addition to price predictions, HOME2U provides immersive 360-degree virtual tours that allow users to explore properties in an engaging and realistic manner. These interactive tours help prospective buyers visualize the space and its potential, offering a comprehensive view that static images alone cannot provide. This functionality is especially beneficial in today’s fast-paced market, where buyers may have limited time to visit multiple properties in person. The ability to conduct virtual walkthroughs right from their devices empowers users to narrow down their choices and make more confident decisions based on their unique preferences and requirements.

To ensure authenticity and build user trust, exclusive administrator control is implemented: all property listings are thoroughly verified by platform administrators before being published, effectively eliminating fake or misleading content that can plague the real estate industry. This rigorous verification process not only fosters trust among users but also enhances the overall quality of listings available on the platform. Furthermore, secure, encrypted communication channels enable safe interactions between buyers, sellers, and agents, protecting sensitive information and fostering a secure environment for negotiation. The commitment to user security is paramount, as HOME2U aims to create a safe digital space where individuals can explore their real estate options without fear of compromise or deceptive practices.

The platform also supports 360-degree property viewing, enabling users to visually explore listings in depth through interactive desktop-based virtual tours helping buyers get a realistic sense of the space without the need for physical visits. This feature is especially useful for out-of-town buyers or those with busy schedules, as it allows them to conduct thorough property research from the comfort of their own homes, saving valuable time and effort in the property search process.

From a technical standpoint, HOME2U is built using modern technologies React.js for a dynamic and interactive frontend that enhances user experience, Node.js for scalable backend operations that ensure reliability and efficiency, MySQL for robust and reliable database management, and AWS for secure and efficient image storage that supports high-resolution visuals. This robust technological foundation ensures that the platform is not only responsive and fast but also capable of handling high volumes of traffic without compromising performance. The use of cloud services like AWS allows for scalable growth, meaning that as the platform expands and attracts more users, it can easily adapt to meet increasing demands, thus ensuring a consistent and high-quality user experience.

Looking ahead, HOME2U plans to expand its capabilities with mobile accessibility, allowing users to search for properties on-the-go through a dedicated mobile application. This will cater to the growing trend of mobile browsing, ensuring that users can stay connected to their property searches wherever they are, whether they are commuting, traveling, or simply enjoying a day out. Additionally, the integration of blockchain technology for secure transactions is on the horizon, promising enhanced security and transparency throughout the entire buying process. By utilizing decentralized ledgers, HOME2U aims to streamline transactions, reduce fraud, and provide an immutable record of ownership, thereby protecting users' investments.

Furthermore, personalized AI-based property recommendations will leverage user data and preferences to suggest listings tailored specifically to individual tastes, making the property search experience even more efficient and enjoyable. By analyzing user behavior and preferences, the platform can refine its offerings, presenting users with properties that align closely with their specific needs and desires. This level of personalization not only enhances user satisfaction but also promotes a more engaging and relevant browsing experience.

By continually innovating and adapting to the needs of its users, HOME2U is committed to remaining at the forefront of the digital real estate landscape, transforming how individuals engage with property exploration and investment. As the platform evolves, it seeks to empower users with the tools and information they need to navigate the real estate market confidently, ensuring that every property search is not just a task but an exciting journey toward finding the perfect place to call home.

1.1 Brief

HOME2U is a next-generation, web-based real estate platform designed to redefine the traditional methods of property buying, selling, and exploration. Addressing long-standing challenges in the real estate industry such as information asymmetry, fraudulent listings, and inefficiencies in property discovery HOME2U offers a suite of integrated, technology-driven solutions tailored to modern user needs. This innovative platform not only streamlines the real estate process but also enhances user experience by providing a more transparent, efficient, and user-friendly interface.

One of HOME2U’s standout features is its AI-powered price prediction model, which uses key property parameters such as area, location, and room count to generate accurate price estimates. This groundbreaking feature empowers users to make more informed decisions, taking into account current market trends and historical data, thereby enhancing transparency in the valuation process. By offering this level of insight, HOME2U allows both buyers and sellers to engage in negotiations with confidence, knowing they are armed with reliable information.

To ensure authenticity and build user trust, exclusive administrator control is implemented: all property listings are thoroughly verified by platform administrators before being published, effectively eliminating fake or misleading content. This rigorous vetting process not only protects users from potential scams but also upholds the integrity of the platform, reinforcing its reputation as a trustworthy real estate resource. Additionally, secure, encrypted communication channels enable safe interactions between buyers, sellers, and agents, protecting sensitive information and fostering a secure environment for negotiation. This focus on security is paramount in today's digital age, where privacy concerns are increasingly prevalent.

The platform also supports 360-degree property viewing, enabling users to visually explore listings in depth through interactive desktop-based virtual tours helping buyers get a realistic sense of the space without the need for physical visits. This immersive experience transforms the way users interact with properties, allowing them to visualize their potential future homes from the comfort of their own devices. The technology behind these virtual tours is designed to simulate an in-person experience, complete with detailed views of each room, outdoor spaces, and even neighborhood amenities, making it an invaluable tool for homebuyers, particularly those relocating from distant areas.

From a technical standpoint, HOME2U is built using modern technologies React.js for a dynamic and interactive frontend, Node.js for scalable backend operations, MySQL for reliable database management, and AWS for secure and efficient image storage. This robust technological foundation not only ensures high performance and reliability but also enables the platform to adapt and evolve with changing user needs and market dynamics. The development team continuously monitors industry trends, integrating new features and enhancements that keep HOME2U at the cutting edge of real estate technology.

Looking ahead, HOME2U plans to expand its capabilities with mobile accessibility, allowing users to access the platform on-the-go, thereby increasing convenience and engagement. Furthermore, the integration of blockchain technology for secure transactions is on the horizon, promising a new level of security and transparency in property dealings, which can significantly reduce the risk of fraud and streamline the closing process. Personalized AI-based property recommendations are also in the works, utilizing advanced algorithms to analyze user preferences and behaviors, ultimately delivering tailored suggestions that align with individual buyer profiles. This commitment to innovation ensures that HOME2U remains at the forefront of the digital real estate space, continually enhancing the user experience and transforming the way people engage with property transactions.

Ultimately, HOME2U is not just a platform; it is a revolution in real estate, combining technology, security, and user-centric design to create an unparalleled experience for buyers and sellers alike. As it continues to grow and evolve, HOME2U is set to redefine the future of property transactions, making them more accessible, efficient, and enjoyable for everyone involved.

1.2 Relevance to Course Modules

This project aligns with multiple Computer Science disciplines, showcasing a comprehensive integration of various technical fields that collectively enhance its functionality and user experience.

- **Software Engineering**

  The project follows the Agile Software Development Life Cycle (SDLC), which is a key framework in contemporary software development. By incorporating iterative planning, requirement gathering, prototyping, and rigorous testing, the project ensures a robust and adaptable end product that can evolve over time. Students are given the opportunity to apply theoretical concepts such as user stories, sprints, and continuous integration to manage and streamline development processes in a real-world context, fostering not only technical skills but also teamwork and communication competencies essential in collaborative environments.

- **Database Systems**

  HOME2U employs MySQL as its primary data storage solution, taking advantage of its powerful capabilities for managing relational databases. The architecture of the database is meticulously designed for both reliability and speed, which is critical for handling complex queries related to property searches and user management. In this context, students learn the importance of database normalization to reduce redundancy, as well as indexing strategies to optimize query performance. Furthermore, transaction management is implemented to ensure data consistency and integrity, which is vital for maintaining user trust and operational excellence.

- **Web Technologies**

  React.js (frontend) and Node.js (backend)

  The choice of React.js for the frontend and Node.js for the backend exemplifies best practices in full-stack development, allowing for a dynamic and responsive user interface coupled with a robust server-side architecture. By utilizing React.js, the project benefits from a component-based architecture that promotes reusability and ease of maintenance, while Node.js facilitates asynchronous data handling, which enhances the overall performance and scalability of the application. This dual approach to web technologies not only reflects industry standards but also prepares students for real-world challenges in web development, enabling them to create seamless and efficient web applications.

- **Human-Computer Interaction**

  The platform is developed with a strong focus on UI/UX principles, which are crucial for creating a positive user experience. This involves ensuring that the interface is not only intuitive and accessible but also visually appealing to engage users effectively. Usability testing and feedback loops are systematically incorporated into the development process to refine user interactions continually. By analyzing user behavior and preferences, the project aims to maximize engagement and satisfaction, ultimately leading to a more successful platform that meets the needs of its diverse user base.

  Beyond these core modules, HOME2U incorporates elements from several other critical areas of technology:

- **Cybersecurity**

  In today's digital landscape, cybersecurity is of paramount importance. The project integrates robust cybersecurity measures that are essential for protecting user data and maintaining platform integrity. This includes implementing encrypted communication channels and secure user authentication protocols. By utilizing SSL/TLS protocols for data transmission, alongside server-side encryption for sensitive content such as images stored on AWS services like Amazon S3, the project ensures that user data is safeguarded against potential breaches, thereby reinforcing trust in the platform's security features.

- **Cloud Computing**

The project leverages cloud-based infrastructure to achieve scalability and flexibility, allowing it to efficiently manage varying user loads without compromising performance. This approach supports horizontal scaling, which is essential in today's dynamic user environments where demand can fluctuate significantly. By adhering to distributed systems and cloud architecture principles, the platform is designed for high availability and resilience, ensuring a seamless user experience across different usage scenarios. This knowledge of cloud computing not only prepares students for the future of technology but also illustrates the practical applications of theoretical concepts in real-world settings.

In summary, the project serves as a practical application of theoretical learning, effectively bridging gaps between different areas of computer science and providing hands-on experience with industry-standard tools and methodologies. By engaging students in a multifaceted project that encompasses a wide array of disciplines, it prepares them for the complexities of modern technology landscapes and equips them with the skills necessary to thrive in their future careers. This holistic approach not only enhances their technical proficiency but also fosters a deeper understanding of how various elements of computer science interconnect and contribute to creating impactful solutions.

1.3 Project Background

Conventional real estate platforms are often plagued by several critical issues: fraudulent or duplicate listings, outdated property information, and lackluster tools for virtual property exploration. These shortcomings not only erode user trust but also create inefficiencies for buyers, sellers, and agents alike, leading to frustration and missed opportunities in an already complex market. As a result, many potential transactions falter before they can even begin, leaving users with a sense of doubt and uncertainty about the integrity of the platform they are using.

- **Admin-controlled listings**

To combat these prevalent issues, HOME2U implements a rigorous verification process for every property submitted to the platform, which is meticulously managed by a dedicated team of administrators. This multi-step process involves cross-referencing listings against public records, conducting background checks on property owners, and employing cutting-edge technologies to identify potentially fraudulent entries. Such diligence ensures that only legitimate, accurate, and current listings are visible to users, drastically reducing scams and misinformation that have historically plagued the real estate market. Consequently, users can navigate the platform with a renewed sense of security, knowing that the properties they view have been thoroughly vetted for authenticity.

- **Immersive virtual tours**

In recognition of the growing demand for innovative digital solutions, HOME2U incorporates advanced 360-degree tour technology to provide users with the opportunity to thoroughly inspect properties online. This exceptional feature is accessible on both desktop and mobile devices, making it convenient for users regardless of their location or preferred browsing method. The visual depth provided by these immersive virtual tours enhances buyer confidence, as prospective buyers can explore properties in detail, examining everything from room layouts to intricate design elements without the need for physical visits. By allowing users to experience properties virtually, HOME2U significantly expedites decision-making, making it easier for buyers to narrow down their choices and move forward with confidence.

- **Advanced search filters**

Recognizing that every buyer has unique preferences and requirements, the platform offers sophisticated search options that enable users to filter properties by a multitude of criteria, including location, area, price range, number of bedrooms, and even specific amenities. This level of personalization streamlines the property discovery process and ensures that users are matched with listings that best suit their individual needs and lifestyle preferences. Recent market research underlines the urgency of these improvements; according to a 2023 survey conducted by Protech Insights, approximately 65% of prospective buyers reported abandoning real estate platforms due to concerns about the authenticity of listings (Real Estate Consumer Behavior Report, 2023). This statistic highlights a widespread lack of trust in existing solutions a gap that HOME2U seeks to fill through its strict verification protocols and user-centric features.

By combining reliable data management, immersive visualization, and robust security, HOME2U sets out to deliver a transparent, efficient, and user-friendly digital marketplace for real estate. These innovations not only address the pressing concerns of contemporary buyers and sellers but also align with the evolving landscape of real estate technology, which demands greater accountability and transparency. Through its commitment to quality and trustworthiness, HOME2U aspires to become a benchmark for future developments in the real estate technology domain, fostering a new era where users can engage with the market confidently and with ease. Ultimately, HOME2U envisions a platform where the complexities of real estate transactions are simplified, and all stakeholders from buyers and sellers to agents can thrive in an ecosystem built on trust, innovation, and mutual success

1.4 Literature Review

A critical examination of existing real estate platforms, such as **Zameen.com** and **OLX.com.pk**, reveals notable limitations in both verification processes and interactive visualization tools. While these platforms have garnered substantial user bases, their reliance on static image galleries and user-generated listings often results in inconsistent data quality and a heightened risk of fraudulent listings. Most notably, these platforms lack admin verification mechanisms and do not offer immersive 360-degree virtual tours, both of which are increasingly demanded by today’s tech savvy real estate consumers. Additionally, an AI-powered price prediction model could further enhance the user experience by providing accurate price estimates, streamlining decision-making for prospective buyers.

Recent research underscores a growing demand for innovative enhancements in digital real estate experiences. For example, several studies advocate for the integration of blockchain technology to facilitate secure property transactions, a feature being considered for future iterations of HOME2U. This technology not only promises to improve transparency and security in transactions but also has the potential to reduce costs associated with traditional real estate processes, thereby making housing more accessible to a broader audience. Moreover, the Journal of Property Tech (2023) highlights the emerging role of artificial intelligence in delivering personalized property recommendations, which can significantly improve user satisfaction and platform engagement. AI algorithms can analyze user preferences, search history, and even social media behavior to curate tailored listings that resonate with individual buyers, ultimately enhancing the likelihood of successful transactions.

In light of these advancements, it is imperative for existing platforms to evolve and adapt to the changing landscape of real estate technology. This could involve not only adopting more sophisticated verification processes to ensure the authenticity of listings but also investing in advanced visualization tools that provide users with a more immersive experience. For instance, incorporating augmented reality (AR) features could allow potential buyers to visualize properties in real-time, superimposing digital enhancements over physical spaces. This would enable users to envision renovations or alterations before making a commitment, thereby increasing their confidence in the purchasing decision.

Furthermore, as the market becomes increasingly competitive, platforms that prioritize user experience and leverage the latest technological innovations will likely emerge as leaders in the industry. By focusing on user-centric design and incorporating feedback mechanisms, platforms can continuously improve their offerings and adapt to the evolving needs of their clientele. As consumers become more informed and discerning, it is crucial for real estate platforms to not only meet but exceed their expectations, ensuring a seamless, efficient, and enjoyable property search experience. Ultimately, the integration of cutting-edge technologies and a commitment to transparency and user satisfaction will be key drivers in transforming the digital real estate landscape for the future.

Recent research underscores a growing demand for innovative enhancements in digital real estate experiences, indicating a shift in how users interact with property markets in the digital age. For example, several studies advocate for the integration of blockchain technology to facilitate secure property transactions, a feature being considered for future iterations of HOME2U. This technology not only promises to streamline the buying and selling processes by eliminating the need for intermediaries but also enhances transparency and reduces the potential for fraud, thus fostering greater trust among users. Furthermore, the Journal of Property Tech (2023) highlights the emerging role of artificial intelligence in delivering personalized property recommendations, which can significantly improve user satisfaction and platform engagement. By leveraging machine learning algorithms, real estate platforms can analyze user preferences, browsing habits, and even social media interactions to curate tailored listings that align with individual tastes and requirements. Such advancements not only make the property search process more efficient but also create a more engaging experience that keeps users returning. As digital real estate technology continues to evolve, the incorporation of virtual reality tours and augmented reality features may further enrich the user experience, allowing potential buyers to explore properties from the comfort of their homes. Collectively, these innovations signal a transformative era in real estate, where technology plays a pivotal role in redefining how properties are marketed, sold, and experienced in the digital landscape. (Journal of Property Tech, 2023)

Comparative analyses published in the Real Estate Tech Journal (2022) demonstrate the tangible impact of advanced visualization: platforms featuring virtual tours retain users approximately 2.5 times longer than those limited to static imagery. This extended engagement is attributed to the richer, more interactive exploration experience provided by immersive media, which allows potential buyers and renters to feel a deeper connection to the spaces they are considering. Rather than simply viewing images that provide a fleeting glimpse of a property, users can navigate through virtual environments, experiencing the layout, lighting, and ambiance as if they were physically present. (Real Estate Tech Journal, 2022)

Furthermore, these immersive experiences enhance user satisfaction and increase the likelihood of conversion, as individuals are more inclined to invest time in properties that they can explore in such a detailed manner. HOME2U's architecture is directly informed by these insights, prioritizing real-time data updates and interactive features to bridge the usability and transparency gaps identified in legacy solutions. By integrating cutting-edge technology such as augmented reality and artificial intelligence, HOME2U not only streamlines the search process but also empowers users with comprehensive information about each property, including neighborhood statistics, market trends, and personalized recommendations based on their preferences.

As the real estate landscape continues to evolve with technological advancements, platforms like HOME2U are setting new standards for user engagement and satisfaction, ensuring that prospective buyers and renters have access to tools that make their journey not just easier, but also more enjoyable and informed. This commitment to innovation reflects a broader shift in the industry towards prioritizing user experience, ultimately transforming the way people interact with real estate and making the process of finding a home more accessible and transparent than ever before.

1.5 Analysis from Literature Review

Synthesizing findings from key studies and industry surveys, several persistent gaps in the online real estate market become evident

- **Trust**

According to the National Realtors Association, a staggering 60% of potential buyers express distrust towards platforms with unverified listings. This skepticism often leads to abandoned transactions and significant reputational damage for these platforms. Trust, therefore, is not merely a preference but a fundamental requirement for success in the online real estate arena. Buyers are increasingly seeking assurance that the listings they are viewing are legitimate and that they can engage in transactions with confidence. Without this trust, platforms risk alienating potential customers, leading to lost revenue and diminished brand loyalty. (National Association of Realtors, 2023)

- **Price Prediction**

To combat this trust issue, HOME2U integrates an innovative AI powered price prediction model that provides users with accurate property price estimates. This advanced feature not only empowers prospective buyers by equipping them with the knowledge needed to make informed decisions but also enhances overall user satisfaction and reinforces trust in the platform's reliability. By utilizing machine learning algorithms that analyze market trends and historical data, HOME2U ensures that users receive up-to-date and realistic pricing, making the daunting task of home buying feel more manageable and less intimidating.

- **Remote Accessibility**

Furthermore, data from TechCrunch (2022) indicates that integrating 360-degree virtual tours significantly increases user engagement, with reports showing an uptick of up to 40%. This enhancement reflects the growing value that users place on remote, in-depth property visualization. In a world where convenience is paramount, the ability to explore properties from the comfort of one’s own home not only saves time but also allows for a more thorough evaluation of potential homes. This feature caters to the needs of busy professionals and long-distance buyers, fundamentally transforming the way properties are marketed and viewed.

- **Search Personalization**

In addition to these features, further analysis reveals that a significant majority 80% of users prefer platforms that offer advanced filtering capabilities. While many platforms traditionally focus on basic attributes like price and location, HOME2U distinguishes itself by simplifying the property search process through the implementation of filters based on specific user preferences such as area, number of rooms, and even unique amenities. This ensures a more streamlined search experience tailored to the most common user needs, ultimately enhancing satisfaction and encouraging repeat usage.

In summary, HOME2U addresses these critical gaps through a sophisticated multi-criteria search engine that empowers users to tailor their property hunt according to nuanced preferences. By integrating advanced filters and prioritizing transparency in listings, HOME2U aligns itself with the latest user expectations and industry best practices. The platform's commitment to building trust, enhancing predictive capabilities, and personalizing searches positions it as a frontrunner in the competitive online real estate market, ensuring that it not only meets but exceeds the demands of today's discerning buyers.

1.6 Methodology and Software Lifecycle for This Project

The development of HOME2U was structured around the Agile methodology, facilitating iterative progress and continuous stakeholder collaboration. This approach not only allowed for flexibility in responding to changing requirements but also fostered a culture of transparency and active engagement among all team members. The software lifecycle was divided into clear phases, each intricately designed to build upon the previous one, ensuring a cohesive and comprehensive development process:

- **Requirement Gathering**

  This initial phase was crucial as it involved engaging with a diverse group of stakeholders, including potential users, business leaders, and technical experts, to define project objectives and user needs. Workshops, interviews, and surveys were utilized to gather insights, which helped in the formulation of a robust set of requirements that accurately reflected the expectations and desires of those who would ultimately interact with the platform.

- **Design**

  Following the requirement gathering, the design phase came into play. Here, the team created detailed wireframes, data models, and system architecture plans that served as blueprints for development. The design process emphasized user-centered principles, ensuring that the interface was intuitive and visually appealing. Collaborative design sessions were held, allowing for feedback loops that further refined the user experience.

- **Development**

  In the development phase, teams focused on constructing both the frontend and backend functionalities, with a keen emphasis on modularity and scalability. This approach ensured that as the platform evolved, new features could be integrated seamlessly without disrupting existing functionalities. Developers employed modern programming languages and frameworks, resulting in a robust and maintainable codebase that supported future enhancements.

- **Testing**

  A rigorous testing phase was implemented, employing both automated and manual tests to ensure feature integrity and system reliability. This included unit tests, integration tests, and user acceptance testing, which collectively aimed to identify and resolve any issues before the platform reached the hands of users. The testing phase was not merely a formality; it was a vital component that underscored the commitment to delivering a high-quality product.

- **Deployment**

  The deployment phase marked a significant milestone as the platform was launched and made available to users. However, this was not the end of the process; real-world performance monitoring allowed the team to gather data on how the system operated under different conditions. Feedback was continuously solicited from users to identify areas for improvement, establishing a loop of ongoing development and refinement.

Each Agile sprint lasted two weeks and incorporated the following core elements, which contributed to the overall success of the project:

- **User Stories**

  Features were prioritized based on user and business value, with specific attention given to the creation of an “Admin Dashboard for Listing Approval.” This feature became central to the platform’s trust model, fostering confidence among users and ensuring that listings were verified and trustworthy before they were made public.

- **CI/CD Pipelines**

  To maintain a rapid development pace while ensuring quality, Continuous Integration/Continuous Deployment (CI/CD) pipelines were established using tools like Jest and Cypress. These automated testing pipelines allowed the team

  to deploy new features rapidly while maintaining system stability, enabling quick iterations based on user feedback.

- **Stakeholder Demos**

  At the end of each sprint, stakeholder demos were conducted, showcasing the latest developments and collecting valuable feedback. These presentations served as a platform for potential users and stakeholders to interact with the product, share their experiences, and suggest enhancements. The insights gained from these sessions were instrumental in refining the platform's UI/UX, ensuring it not only met but exceeded evolving expectations.

In summary, the methodology and software lifecycle employed in the development of HOME2U were meticulously designed to create a dynamic, user-focused platform that could adapt to the needs of its users, while also fostering a collaborative environment that valued stakeholder input at every stage of the process. This strategic approach laid a strong foundation for the successful launch and ongoing evolution of HOME2U, assuring its relevance in a rapidly changing digital landscape

1.7 Rationale behind Selected Methodology

Agile was selected for its inherent flexibility and emphasis on stakeholder involvement, both of which proved indispensable throughout the HOME2U project. The iterative nature of Agile allowed the team to adapt quickly to changing requirements, incorporating feedback and addressing unforeseen challenges in near real-time. This adaptability became particularly valuable when initial user testing indicated that a model allowing unrestricted, user-generated listings posed significant risks to data quality and platform credibility. By pivoting to an admin curated system mid-development without derailing the overall project timeline the team demonstrated the practical advantages of Agile over more rigid methodologies.

The decision to transition to an admin-curated model was not made lightly; it involved comprehensive discussions among team members, user representatives, and stakeholders to ensure that all perspectives were considered. This collaborative approach is one of the hallmarks of Agile, as it fosters an environment where continuous communication and feedback are prioritized. The team held regular sprint reviews, which not only facilitated transparency but also empowered stakeholders to voice their concerns and suggestions, ultimately leading to a more robust solution.

Furthermore, Agile's framework allowed the team to break down the project into manageable sprints, each focusing on specific functionalities and improvements. This incremental progress meant that the team could quickly validate assumptions, test new features, and gauge user reactions, ensuring that the final product was finely tuned to meet user needs and expectations. Each iteration built upon the last, creating a sense of momentum and accomplishment that kept morale high and the project on track.

The incorporation of user feedback was especially critical when navigating the complexities of online real estate transactions, where trust is paramount. By engaging users throughout the development process, the team was able to identify potential pain points and address them proactively. This proactive stance not only mitigated risks but also enhanced user satisfaction, as individuals felt their insights were valued and directly influenced the platform's evolution.

Ultimately, Agile's capacity for rapid iteration, ongoing feedback integration, and swift response to emergent issues ensured that HOME2U remained aligned with its primary mission: fostering trust and transparency in online real estate transactions. The methodology's inherent focus on user-centered design proved instrumental in building a platform that not only meets functional requirements but also resonates deeply with users, establishing a strong foundation for lasting relationships in the marketplace. As HOME2U continues to evolve, the lessons learned from employing Agile will undoubtedly serve as a guiding framework for future projects, reinforcing the importance of adaptability and stakeholder collaboration in an ever-changing digital landscape.

Chapter 2

Problem Definition

2 Problem Definition

This section delves into the fundamental challenges faced by users of traditional real estate platforms and articulates how HOME2U is strategically designed to overcome them. It not only identifies the pain points plaguing buyers and sellers but also clarifies the expected deliverables, outlines stakeholder needs, and positions HOME2U within the competitive landscape. By analyzing both the shortcomings of current solutions and the unique features of HOME2U, this section highlights the project's distinct value proposition, ultimately aiming to revolutionize the way property transactions are conducted.

2.1 Problem Statement

The real estate market, as facilitated by conventional digital platforms, is fraught with obstacles that impede transparent, efficient, and secure property transactions. Buyers are particularly vulnerable to several recurring issues that not only complicate their search for the perfect home but also create an environment ripe for exploitation:

- **Unverified Listings and Fraud**

  Many platforms permit open, user-generated listings without rigorous checks. This creates opportunities for fraudulent activity, with buyers often encountering fake, duplicated, or misleading property information that can lead to financial loss and wasted time. The prevalence of such scams erodes trust in the entire market and discourages potential buyers from engaging fully, as they remain skeptical of the authenticity of the listings they encounter**.** (Protech Insights, 2023)

- **Price Prediction**

  Traditional platforms often lack tools for estimating property prices, leading to uncertainty about fair market value. Buyers, left to rely on outdated comparable or gut instincts, are at a disadvantage when negotiating. HOME2U addresses this gap with an AI-powered price prediction model, utilizing advanced algorithms and vast datasets to help users make informed decisions by providing accurate price estimates based on current market trends and historical data points, thereby empowering them to negotiate with confidence and clarity**.**

- **Inadequate Property Visualization**

  The reliance on static images fails to convey the true condition and spatial dynamics of properties. Buyers are left to make critical decisions with incomplete information, increasing uncertainty and decreasing satisfaction. This lack of immersive experiences can lead to buyer's remorse, where clients find themselves disappointed upon visiting properties that appeared more appealing online. HOME2U aims to combat this issue by integrating virtual reality and 3D walkthroughs, enabling potential buyers to explore homes in a more interactive and informative manner.

- **Inefficient communication**

  Communication channels on many platforms are either insecure or poorly managed, causing delays in responses, confusion, and a lack of accountability. Buyers can struggle to get timely, accurate answers, which slows the decision-making process and often results in missed opportunities in a competitive market. HOME2U's innovative approach to communication ensures that inquiries are handled swiftly and securely, minimizing the risk of miscommunication and enhancing the overall user experience.

Sellers, on the other hand, face challenges such as decreased visibility for their listings due to marketplace clutter and inconsistent moderation. Their outreach is often limited, and genuine inquiries can be lost among spam or unqualified leads, leading to frustration and diminished motivation to list properties. HOME2U addresses these issues by implementing a system of admin-mediated communication, where every inquiry is routed through a controlled process. Prototype testing has shown that this approach reduces response times by an estimated 30%, ensuring buyers receive prompt, reliable information while sellers benefit from higher-quality leads and greater exposure.

By systematically addressing these multifaceted challenges, HOME2U not only enhances user satisfaction for both buyers and sellers but also aims to establish a more trustworthy and efficient real estate ecosystem. In doing so, it positions itself as a leader in the market, setting new standards for how real estate transactions should be conducted in the digital age. The commitment to innovation and user-centric design reflects HOME2U's mission to redefine the property search experience, making it accessible, transparent, and ultimately rewarding for everyone involved.

2.2 Deliverable and Development Requirements

To fulfill its mission, HOME2U will deliver a comprehensive suite of both functional and nonfunctional features, along with a strategic roadmap for future enhancements aimed at providing an exceptional user experience and maintaining long-term relevance in the ever-evolving real estate market:

2.2.1 Functional Requirements

- **Hero Section**
  - The website will prominently display an engaging banner that captures the essence of HOME2U, visually representing its commitment to connecting users with their ideal properties.
  - A sophisticated filter feature will be incorporated, empowering users to seamlessly refine their property searches based on various criteria such as location, area, number of rooms, and other essential attributes. This feature aims to enhance user engagement by facilitating a personalized browsing experience.
- **Property Section**
  - Admin users will possess the capability to create, read, update, and delete property listings with ease, ensuring that the platform remains up-to-date and relevant. o Normal users will have the ability to view property listings, which can either be filtered according to their selected criteria or displayed in entirety for broader exploration.
  - Each property will be showcased with critical information, including high-quality images, pricing, precise location details, and additional relevant specifications such as square footage, amenities, and neighborhood insights, thereby enriching the user's understanding of each listing.
- **About Section**
  - This section will serve to provide comprehensive company information, detailing the origins of HOME2U, its mission, and the various services offered to users, fostering a sense of trust and transparency.
- **Services Section**

  The platform will feature hard-coded content that eloquently describes the array of services offered, such as property listing management, real estate consultancy, and market analysis, thus clearly articulating the value proposition to potential users.

- **Contact Section**
    - Users will be able to easily send inquiries or feedback via a well-designed contact form that directs emails to the platform's official email address, ensuring effective communication and support.
- **Review Section**
    - This area will display hard-coded reviews and testimonials from alumni or users who have engaged with the platform, providing social proof and building credibility for HOME2U among prospective users.
- **AI Price Prediction Model**
    - A floating button will be integrated into the interface, which, when clicked, opens a user-friendly form designed to collect property details from users interested in price estimation.
    - This form will enable users to input essential information such as location, size, number of rooms, and other relevant factors, allowing them to predict the price of a property based on the data entered.
    - The AI model will utilize sophisticated algorithms to analyze the input data and generate a predicted price, enhancing the user experience by providing valuable insights into property valuation.
- **AWS S3 Bucket for Image Storage**
    - Property images will be uploaded and securely stored in an AWS S3 bucket, safeguarding them against unauthorized access while ensuring robust performance.

    - The platform will implement measures to ensure that these images are easily accessible to users, with stringent security protocols and permissions in place to protect the integrity of the content.

2.2.2 Non-Functional Requirements

- **Performance**
  - The website is designed to load within an optimal timeframe of 3 seconds, thereby ensuring a smooth and uninterrupted user experience, which is critical in retaining user interest and engagement.
  - Property images, particularly those showcasing listings, will be meticulously optimized to ensure rapid loading times without sacrificing visual quality, enhancing the overall aesthetic appeal of the platform.
- **Scalability**
  - HOME2U is designed with scalability in mind, enabling the platform to accommodate increasing traffic and a growing number of property listings as its user base expands.
  - The AI model will also be built to scale efficiently, ensuring that it can handle increased usage without compromising performance or accuracy.
- **Security**
  - Ensuring secure communication between users and the platform is paramount; therefore, SSL/TLS encryption will be implemented for all data transmissions to protect sensitive information. (OWASP Security Guidelines, 2023)
  - User data and property details will be securely stored and processed, adhering to best practices in data protection and privacy.
  - The platform will implement robust authentication measures for admin users, allowing them to manage property listings safely and effectively.
  - The use of an AWS S3 bucket will be accompanied by appropriate access control policies designed to safeguard image storage against unauthorized access.
- **Usability**
  - The platform will prioritize user-friendliness, featuring an intuitive navigation structure and interface that facilitates a seamless journey for users.
  - A responsive design will be implemented, particularly for the hero section and filter functionality, ensuring that users enjoy a consistent experience across various devices, from desktops to mobile phones.
  - The AI price prediction form will be streamlined for simplicity and ease of use, allowing users to input data with minimal effort.

- **Availability**
    - HOME2U will strive to maintain high availability, ensuring that the website is operational 24/7 with minimal downtime for maintenance or updates, thereby providing users with continuous access to property information.
    - The AI model and property pages will be consistently accessible, reinforcing the platform's reliability and user trust.
- **Compatibility**
    - The platform will be designed to function seamlessly across modern web browsers, including Chrome, Firefox, Safari, and Edge, ensuring that users can access HOME2U regardless of their browser preference.
    - Ensure the website is responsive and works well on both desktop and mobile devices, despite it being primarily designed for desktop.
- **Maintainability**
    - The architecture of the platform will be established in a way that allows for straightforward updates and bug fixes, employing React components for modularity and ease of maintenance.
    - Thorough documentation will be maintained for the codebase to support future updates, facilitate team collaboration, and ensure clarity in development processes.
- **Reliability**
    - The AI model will be designed to consistently provide accurate price predictions based on the data provided by users, thereby establishing trust in its functionality.
    - Property listings will be updated and displayed correctly, free from errors, thereby enhancing user confidence in the platform's accuracy.
- **Future Deliverables**
    - Plans are in place to extend the HOME2U experience to mobile applications, targeting both iOS and Android devices, which will further expand the platform’s reach and provide users with convenient access to listings on-the-go.
    - An AI-powered chatbot will be introduced as part of future enhancements, designed to handle frequently asked questions, automate customer support, and alleviate the administrative workload, ultimately enhancing the overall user experience by providing immediate assistance and information.

These comprehensive requirements ensure that HOME2U is not only robust in its initial incarnation but also adaptable and poised for future growth, responding effectively to the evolving needs of stakeholders and users alike.

2.3 Current System

The prevailing model among Pakistani real estate platforms such as OLX and Property24 relies heavily on user-generated listings with minimal oversight, which inherently leads to a myriad of challenges that can frustrate potential buyers and sellers alike. This approach results in frequent data inconsistencies, outdated or duplicate property entries, and a lack of standardized, high quality listing information that users have come to expect in today's digital marketplace. As a consequence, prospective homeowners or investors often find themselves sifting through a convoluted array of listings, many of which may not even be available anymore, making the entire property search process not only tedious but also time-consuming.

Moreover, virtual tours and immersive visualization tools technologies that have revolutionized the way real estate is showcased are rare or entirely absent on these platforms, severely limiting users' ability to assess properties remotely. This lack of advanced visual tools means that buyers cannot fully appreciate the features of a property until they visit in person, which can lead to wasted trips and disappointment if the property does not meet their expectations.

In addition to these operational inefficiencies, a comprehensive competitive analysis revealed a significant security gap: none of the current top five platforms in the region offer end-to-end encrypted messaging. This glaring oversight exposes users to privacy risks during negotiations, where sensitive information such as financial details and personal contact information may be at risk of interception by malicious actors. Such vulnerabilities can undermine trust in the platform, as users may hesitate to engage fully in the buying or selling process if they feel their information is not secure.

By addressing these deficiencies particularly through the introduction of admin-verified listings to ensure authenticity, immersive virtual tours that allow potential buyers to explore properties from the comfort of their homes, and secure communication channels that facilitate safe negotiations HOME2U is positioned to set a new benchmark for digital trust, user engagement, and market efficiency in the real estate domain.

In fact, a competitive analysis showed that none of the top five Pakistani real estate platforms offer end-to-end encrypted messaging, a critical gap that HOME2U fills effectively to protect sensitive buyer-seller negotiations. By prioritizing user security and enhancing the overall user experience with innovative features, HOME2U not only aims to fill these gaps but also strives to cultivate a community of informed and empowered users. This strategic shift not only addresses the current shortcomings of existing platforms but also fosters a culture of transparency and reliability in the real estate market, ultimately leading to a more streamlined and satisfying experience for all involved. As the digital landscape continues to evolve, HOME2U stands ready to lead the charge, ensuring that every transaction is not only efficient but also secure and trustworthy, thus redefining the future of real estate in Pakistan

Chapter 3
Requirement Analysis

3 Requirement Analysis

This section systematically dissects the functional and non-functional requirements necessary for the successful development and operation of HOME2U. It combines visual modeling (use case diagrams), user stories, and risk assessments to capture both behavioral and technical specifications from the perspectives of diverse platform actors (admins, buyers, visitors).

3.1 Use Cases Diagram(s)

Use case diagrams will be constructed to visually map out the interactions between key actors Admin, Buyer, and Visitor and the HOME2U system. These diagrams will identify and illustrate the primary use cases, such as listing approval, property browsing, inquiry submission, and admin intervention, providing a clear overview of how users engage with and benefit from the platform's features.

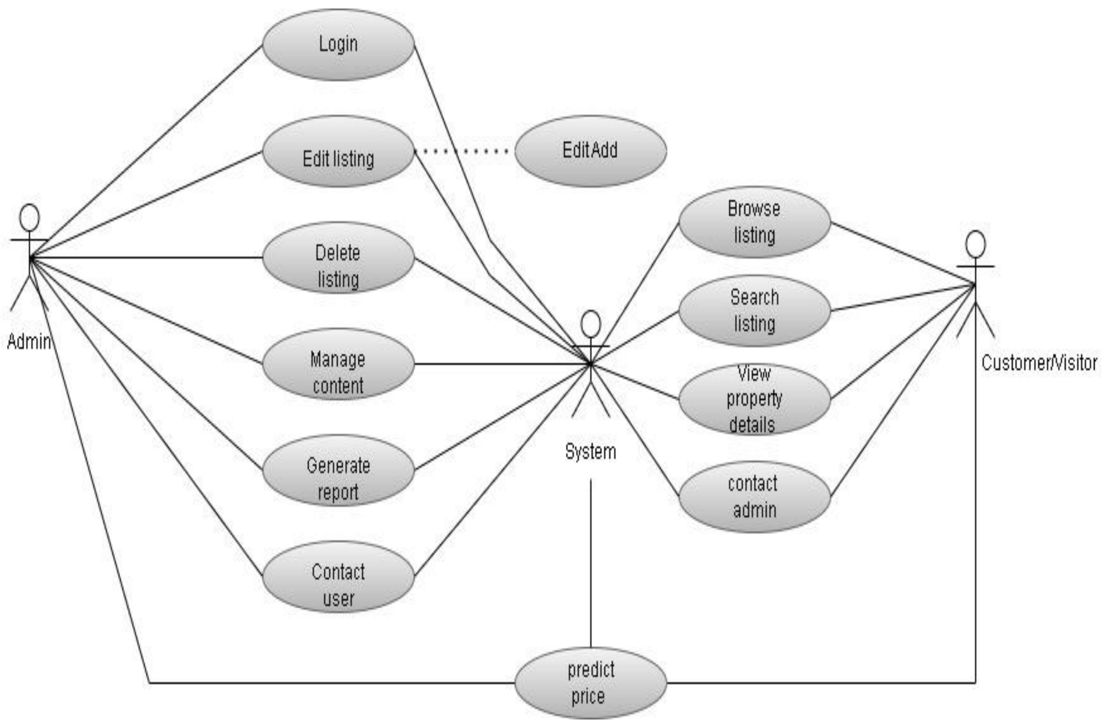

Figure 1 Use Case of Real Estate System

3.1.1 Use Case for Customers

- **Searching for Properties**

  Users can search for properties based on criteria such as location, price range, property type, and amenities.

- **Predicting Property Prices**

  Users can get estimated property prices using intelligent algorithms based on location, size, and market trends.

- **Viewing Property Details**

  Users can view detailed information about individual properties, including photos, descriptions, features, and contact details for the listing agent.

- **Contacting Admin**

  Users can initiate contact with Admin through messaging or contact forms.

3.2 Detailed Use Case

Detailed descriptions of each use case will be provided, outlining the steps involved and the interactions between the user and the system. Use case descriptions will include reconditions, postconditions, main flow of events, alternate flows, and exception handling scenarios.

Table 1 Login

Use Case ID	01
Use Case Name	Login
Actors	Admin
Description	Allows an actor to log in to the system.
Trigger	The admin wants to access their account.
Pre-Condition	The admin must have a registered mail.
Post-Condition	The admin is logged into the system.

Normal Flow	1. The user accesses the login page. 2. The user enters their email and password. 3. The system validates the credentials. 4. The user is granted access to their account.
Exceptions	E1: If the credentials are invalid, the system displays an error message.
Business Rules	BR1: The system must encrypt passwords. BR2: The system must log failed login attempts.

Table 2 Edit Listing

Use Case ID	02
Use Case Name	Edit Listing
Actors	Admin
Description	Allows the admin to manage property.
Trigger	The admin wants to update a listing.
Pre-Condition	The listing must exist in the system.
Post-Condition	The listing is updated with new information.
Normal Flow	1. The admin navigates to the listing management page. 2. The admin selects a listing to edit. 3. The admin modifies the necessary details. 4. The system saves the updated listing. 5. A confirmation message is displayed.
Exceptions	E1: If the listing does not exist, the system displays an error message.
Business Rules	BR1: Listings must contain valid and complete details.

Table 3 Delete Listing

Use Case ID	03
Use Case Name	Delete Listing

Actors	Admin
Description	Allows the admin to remove a property listing from the system.
Trigger	The admin wants to remove a listing.
Pre-Condition	The listing must exist in the system.
Post-Condition	The listing is deleted from the system.
Normal Flow	1. The admin navigates to the listing management page. 2. The admin selects a listing to delete. 3. The system prompts for confirmation. 4. If confirmed, the system removes the listing. A confirmation message is displayed.
Exceptions	E1: If the listing does not exist, the system displays an error message.
Business Rules	BR1: Deleted listings cannot be recovered.

Table 4 Browse Listing

Use Case ID	04
Use Case Name	Browse Listings
Actors	Admin, Customer
Description	Allows a customer to browse available property listings.
Trigger	A user wants to explore properties.
Pre-Condition	The system must have active listings.
Post-Condition	The user can view available properties.
Normal Flow	1. The customer accesses the browsing page. 2. The system displays available listings. 3. The customer scrolls through and views property details.
Exceptions	E1: If no listings are available, the system shows an appropriate message.
Business Rules	BR1: Only active listings are shown to customers.

Table 5 Search Listing

Use Case ID	05
Use Case Name	Search Listings
Actors	Customer, Visitor
Description	Allows a customer to search for specific property listings.
Trigger	A user wants to find a property based on certain criteria.
Pre-Condition	The system must have searchable listings.
Post-Condition	The user sees relevant search results.
Normal Flow	1. The customer accesses the search page. 2. The customer enters search criteria (e.g., location, price, type). 3. The system retrieves and displays matching listings.
Exceptions	E1: If no listings match, the system displays a message.
Business Rules	BR1: Search results must be relevant to the input criteria.

Table 6 View Property Detail

Use Case ID	06
Use Case Name	View Property Details
Actors	Customer, Visitor
Description	Allows a customer to view details of a selected property.
Trigger	A user selects a property from the listing.
Pre-Condition	The property must exist in the system.
Post-Condition	The user sees full details of the property.
Normal Flow	1. The customer selects a property. 2. The system displays full property details, including price, features, and images.
Exceptions	E1: If the property does not exist, an error message is displayed.

Business Rules	BR1: Property details must be accurate and up to date.

Table 7 Contact Admin

Use Case ID	07
Use Case Name	Contact Admin
Actors	Customer, Visitor
Description	Allows a customer to contact the site administrator for assistance.
Trigger	A user needs help or has an issue.
Pre-Condition	The admin contact option must be available.
Post-Condition	The user successfully sends a message to the admin.
Normal Flow	1. The customer navigates to the contact admin page. 2. The customer enters their query and submits the form. 3. The system forwards the query to the admin. 4. The admin responds through the provided contact method.
Exceptions	E1: If the admin contact option is disabled, the system displays an error.
Business Rules	BR1: Admin contact requests must be handled within a reasonable time.

Table 8 Price Prediction

Use Case ID	08
Use Case Name	Price Prediction (Admin)
Actors	Admin
Description	Allows the admin to manage and update the price prediction algorit using latest market data
Trigger	The admin wants to update or refine the prediction model.
Pre-Condition	The prediction system must be implemented and accessible.
Post-Condition	The prediction model is updated or retrained.

Normal Flow	1. Admin navigates to the model management section. 2. Admin uɼ new data or adjusts parameters. 3. System retrains the model using updated data. 4. System confirms the model has been updated.
Exceptions	E1: If data is invalid or corrupt, an error message is displayed.
Business Rules	BR1: Only verified admins can access and modify the model.

Table 9 Price Prediction(Customer)

Use Case ID	09
Use Case Name	Price Prediction (customer/visitor)
Actors	Customer/visitor
Description	Allows users to get an estimated price of a property based on input parameters.
Trigger	A user wants to estimate the market value of a property.
Pre-Condition	The user provides sufficient input data (e.g., location, size, property type).
Post-Condition	The system displays an estimated price based on the prediction model.
Normal Flow	1. User opens the price prediction tool. 2. User enters property details (location, size, features, etc.). 3. System processes the input using the prediction model. 4. Estimated price is shown to the user.
Exceptions	E1: If input is incomplete or invalid, system show message (error)
Business Rules	BR2: Estimated prices are for guidance only and not final offers.

3.3 Functional Requirements

Functional requirements include features for both customers and admins. These features ensure a smooth user experience and efficient platform management.

3.3.1 Customer's Module

- **Property Search**

  Users are empowered to conduct comprehensive property searches using a variety of criteria, including but not limited to location, price range, property type, number of bedrooms, and specific amenities. The search functionality incorporates advanced filtering options to refine results, ensuring users quickly find properties that align with their preferences. Users can also save their search preferences and set up custom alerts, receiving notifications when new listings that match their criteria become available maximizing efficiency and keeping them informed of market updates.

- **Price Prediction**

  Users can access an intelligent price prediction tool that provides real-time property value estimates based on key inputs such as location, property size, type, and current market trends. Powered by machine learning algorithms, the system analyzes historical data and regional pricing fluctuations to generate accurate, data-driven predictions. This feature helps buyers and sellers understand fair market value, set realistic expectations, and make confident, informed decisions during negotiations.

- **View Property Details**

  Each property listing provides an in-depth view, featuring high-resolution images, 360degree virtual tours, detailed textual descriptions, and a breakdown of features such as square footage, year built, neighborhood data, energy certifications, and recent renovations. Transparent pricing information is always presented, alongside historical price trends when available, empowering buyers to make informed decisions.

- **Contact Admin**

  To promote secure and streamlined communication, users can easily contact the admin team through integrated messaging systems or dynamic contact forms. All inquiries are logged and routed directly to authorized personnel, ensuring a prompt, professional response. Users can track their communication history and receive real-time updates on the status of their inquiries.

3.3.2 Admin Modules

- **Property Management**

  Administrators have access to a robust dashboard for managing property listings. This includes the ability to add new properties with multimedia content, edit listing details for accuracy, and remove properties that are sold, delisted, or outdated. The system maintains a complete audit trail, recording every change with time stamps and admin identification to ensure transparency and accountability.

- **Reporting and Analytics**

  The admin interface incorporates powerful analytics tools and customizable reporting features. Admins can view metrics such as user activity (searches, views, inquiries), property performance (engagement rates, time on market), and broader search trends. These insights inform strategic decisions such as optimizing listing visibility, identifying user demand shifts, and refining the platform's feature set for maximum impact.

3.4 Non-Functional Requirements

Non-Functional Requirements of the HOME2U platform focus on usability, performance, security, portability, and reliability.

3.4.1 Usability

- The platform's user interface is meticulously designed for clarity, simplicity, and intuitive navigation. Clear labeling, logical menu structures, and consistent visual elements guarantee that users regardless of technical proficiency can easily access all features and information.
- The website employs responsive design principles, automatically adapting layouts and controls for optimal usability on desktops, laptops, tablets, and smartphones. Accessibility best practices are followed to accommodate users with disabilities.
- Users receive immediate feedback on their actions, such as loading states, confirmation messages for successful operations, and clear error notifications with guidance for resolution.

3.4.2 Performance

- HOME2U is engineered for performance, with code and assets optimized to deliver near-instant page loads and seamless navigation. Key user actions (like searching and messaging) are designed to execute within sub-second response times under typical conditions.
- Rigorous load testing simulates peak user activity to identify and address any performance bottlenecks, ensuring the platform remains reliable and responsive as user volume grows.

3.4.3 Security

- The platform enforces robust security protocols at every level. Admin authentication utilizes secure methods, and all sensitive data including admin credentials and personal information is encrypted both at rest and in transit.
- Regular penetration testing and security audits are conducted to identify and mitigate potential vulnerabilities, maintaining user trust and regulatory compliance.

3.4.4 Portability

- HOME2U is compatible with all major web browsers (e.g., Chrome, Firefox, Safari, Edge) and operating systems (Windows, macOS, Linux, iOS, Android), ensuring a consistent experience for every user.
- Extensive device testing is performed to verify functionality and appearance on a broad spectrum of hardware, from large desktop monitors to compact smartphone screens.

  The site's adaptive design guarantees usability regardless of screen size or orientation.

3.4.5 Reliability

- System uptime is prioritized, with monitoring and redundancy measures in place to minimize downtime. Automated failover and backup systems support business continuity in the event of hardware or software failures.
- Data integrity is safeguarded through scheduled backups and disaster recovery protocols, protecting against both accidental data loss and malicious threats.

Regular backups and recovery plans should be maintained to protect data from loss or security threats.

Chapter 4
Design and Architecture

4 Design and Architecture

This section presents the architectural foundation of HOME2U, detailing the system's core components, technological stack, data flow, and scalability strategies. Visual artifacts such as entity-relationship diagrams and process flows are used to articulate the platform's structure and facilitate future enhancements, ensuring that stakeholders have a clear understanding of the system's capabilities and design principles.

4.1 System Architecture

HOME2U operates on a modern client-server architecture, designed to optimize both user experience and system performance. The front end is a desktop-only website developed using React, providing users with interactive property search, browsing, and communication features to enhance engagement. The interface is visually appealing and highly functional, offering smooth navigation and fast access to property listings. The backend server is robust, handling essential business logic, database operations, and user authentication ensuring the integrity and security of user data. All data exchanges occur over secure, encrypted channels, protecting sensitive information from unauthorized access.

The architectural design process typically includes these seven phases

- **Pre-Design**
    - This initial stage focuses on collecting detailed client requirements, analyzing the target market, and establishing clear project goals. By engaging with stakeholders and potential users, the design team can gather insights that will inform every subsequent phase.
- **Schematic Design**
    - During this phase, conceptual sketches and wireframes are created to define the platform's structure, user flows, and feature set. This visual representation helps in aligning the team's understanding of the project and serves as a foundation for further development.
- **Design Development**
    - Building upon the initial concepts, this phase involves elaborating on the designs, selecting appropriate technologies, and specifying integration points and APIs that will connect various components of the system. Careful consideration is given to

ensure that each technology choice aligns with the project's goals and performance expectations.

- **Construction Documents**
    - Detailed technical documentation is produced for developers, including comprehensive wireframes, data schemas, and API specifications. These documents serve as a blueprint for the development team, ensuring that everyone is on the same page regarding the technical requirements and expected outcomes.
- **Bidding**
    - In this phase, the project team engages with development partners and vendors to estimate costs and select implementation teams. This process involves careful evaluation of proposals to ensure that the chosen partners have the necessary expertise and resources to bring the project to life.
- **Construction Administration**
    - Overseeing the development process, this phase includes conducting code reviews and managing deployments to ensure fidelity to the design specifications. Regular check-ins and progress assessments help to address any issues that arise, maintaining the project's timeline and quality standards.
- **Post-Construction**
    - After the initial launch, ongoing support and monitoring are implemented, along with iterative improvement cycles based on user feedback and analytics. This phase is crucial for adapting the platform to meet changing user needs and for continuously enhancing the overall user experience.

Each phase builds on the last, ensuring a cohesive, well-executed project from concept to completion, and allowing for flexibility in responding to any challenges or new opportunities that may arise during the development process

4.2 Data Representation

4.2.1 Database Schema

The HOME2U database schema is constructed with both flexibility and integrity in mind, leveraging relational principles to capture all relevant entities and interactions effectively. Core tables are designed to ensure that data is organized and easily accessible, facilitating efficient queries and updates. Key tables include

- **Users**

  This table stores comprehensive user profiles, authentication data, and saved preferences, allowing for personalized experiences and streamlined logins.

- **Properties**

  This table contains detailed property listings, multimedia content, and metadata that provide potential buyers or renters with a rich understanding of each listing.

4.2.2 Relationships

The schema establishes key relationships among the entities

- **One-to-many**

  An Admin can manage multiple Properties, reflecting the administrative structure and responsibilities within the platform.

- **Many-to-one**

  Visitors can view multiple Properties, with each view recorded for analytics this data is invaluable for understanding user behavior and preferences.

All sensitive operations are logged, with every price change, status update, or property edit timestamped and mapped to an admin identity, ensuring transparency and accountability.

4.3 Design Models

A comprehensive Entity-Relationship (ER) diagram depicts the platform's main entities Admin, Property, Visitor and their interrelations in a visually coherent manner. Each entity captures a rich set of attributes, such as Property_ID, Price, and Location, enabling robust querying and analytics that are vital for operational insights and strategic decision-making. Visitor activity is anonymized and aggregated, allowing the platform to inform continuous improvements while preserving user privacy. This approach to analytics not only helps admins optimize property listings and search features but also ensures that the platform remains responsive to the evolving needs of its user base without compromising confidentiality.

4.3.1 ER Diagram

- Entities: Admin, Property, Visitor and model.

- Attributes: Property_ID, Price, Location.

The **Visitor** entity captures anonymized analytics (e.g., search trends) to help admins optimize listings without compromising privacy.

4.4 Process Flow/Representation

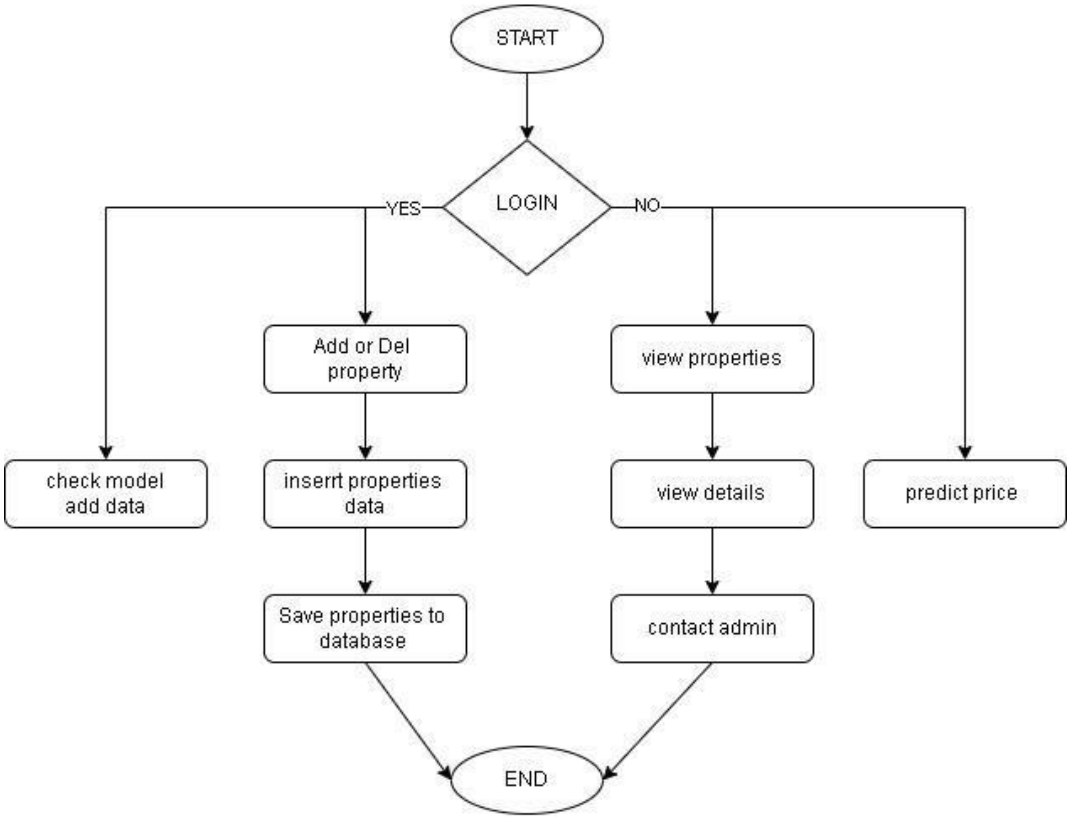

Figure 2 Flow Diagram of Real Estate System

4.4.1 Description

- **Start**
    - The process begins when a user navigates to the property management system interface.
    - This is the entry point for all users admins and general visitors alike.
- **Login Decision (Yes/No)**
    - The system checks if the user is an admin. Only admins can log in.

- **Yes:** If the user is an admin, they proceed to login.
- **No:** Non-admin users bypass login and continue as guests to view properties or predict prices.

- **Admin Login**
  - Admins enter their credentials (specific mail). After successful authentication, they access administrative functionalities like managing property listings and updating data models.
- **Check Model / Add Data**
  - Admins can verify the prediction model and feed new data into it to keep it accurate and up to date.
- **Add or Delete Property**
  - Admins can add new property listings or delete outdated ones as part of their management tasks.
- **Insert Property Data**
  - After choosing to add a property, the admin inputs necessary information such as price, location, size, features, and images.
- **Save Property to Database**
  - Once data entry is complete, the system saves the information into the database, making it available for public viewing.
- **View Properties (user Access)**
  - Non-logged-in users (guests) can browse all publicly available property listings.
- **View Details (user)**
  - A guest can select a specific property to see detailed information, including features, price, and media**.**
- **Contact Admin (user)**
  - If interested, guests can reach out to the admin for inquiries via a contact form or provided contact details.
- **Predict Price (user)**
  - Guests can use a built-in feature to estimate property prices based on selected criteria, without logging in.

- **End**
  - After completing their actions whether managing properties (admin) or browsing/searching (guest) users exit the system.

This diagram represents a straightforward workflow for managing property listings, including adding, editing, viewing, and contacting sellers, with user authentication (login/logout) integrated into the process. It highlights the user-centric design of the system, making it accessible and efficient for all users, regardless of their level of experience with technology

4.5 Design Models

In this section, it is essential to include design models that provide a clear and comprehensive understanding of the system's structure and functionality. These models serve as blueprints for how the system operates, illustrating the relationships between various components and outlining the user experience in a visual format. By employing design models such as flowcharts, entity relationship diagrams, and user interface mockups, stakeholders can gain insights into the system's architecture and functionality, facilitating better communication and collaboration throughout the development process. These models will not only clarify the workflow but also highlight potential areas for improvement, ensuring that the final product is user-friendly, efficient, and aligned with the needs of its target audience.

4.5.1 Sequence Diagram

This is the following Sequence Diagram of our project

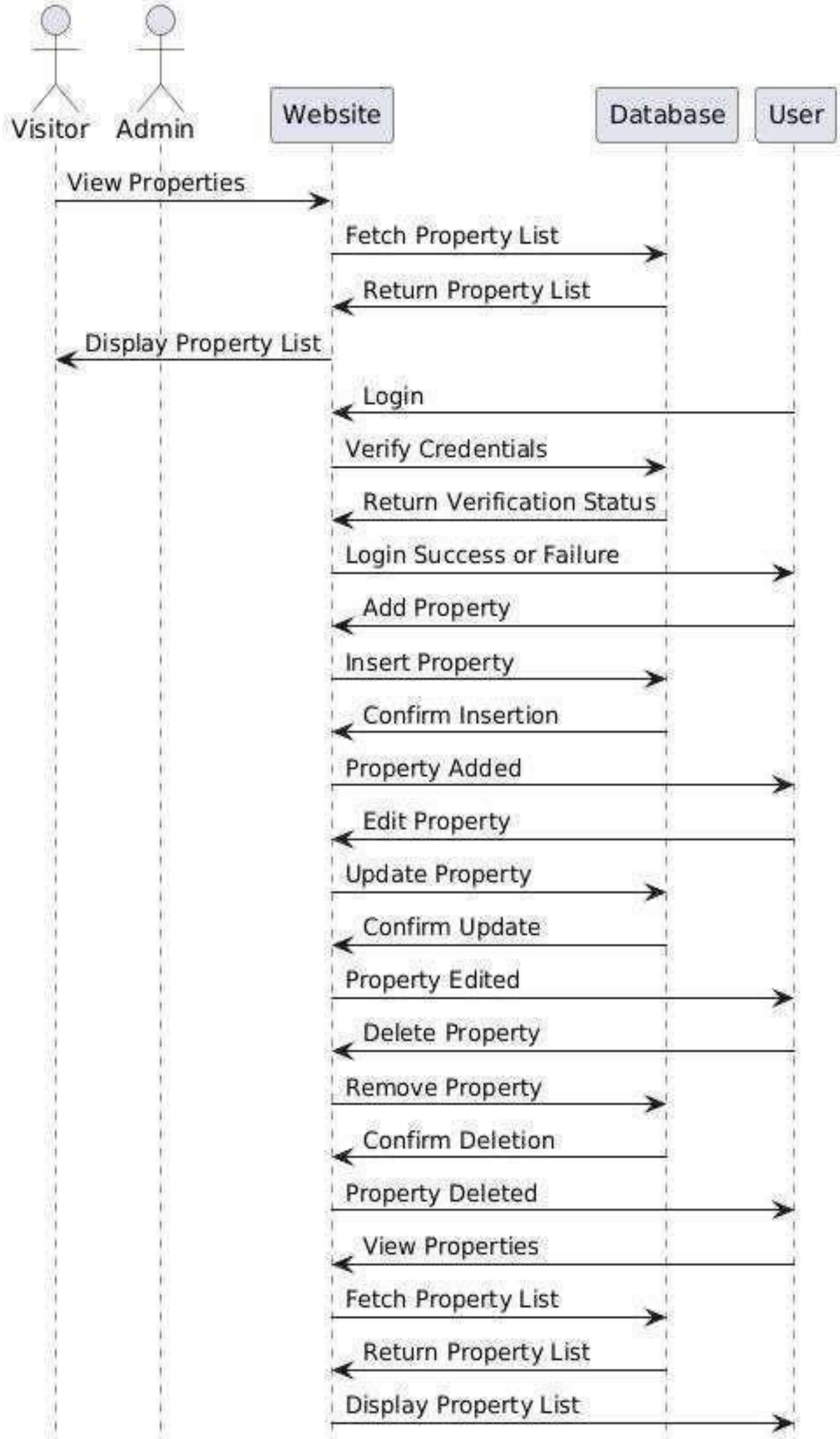

Figure 3 Sequence Diagram of Real Estate System

4.5.2 Description

- **Visitor Perspective**

In the visitor perspective of the system, users are granted the opportunity to explore a wide array of properties tailored to their preferences and needs

- **View Properties**

  Visitors can easily browse through the extensive list of available properties, which may include residential homes, commercial spaces, land for development, and vacation rentals. The system efficiently fetches the property list from the database, ensuring that visitors are presented with the most current and relevant listings available in real-time. This seamless integration with the database not only enhances user experience but also ensures that visitors can make informed decisions based on up-to-date information

- **Contact Admin via Contact Form**

  Visitors have the option to easily contact the admin for any property-related inquiries. A user-friendly **Contact Form** is available on the platform, allowing users to enter essential information such as their name, email, phone number, and message. This feature provides a simple and direct way for visitors to reach out to the admin with questions, requests for more information, or to express interest in a property.

- **Price Prediction Model**

  Visitors have the option to estimate the price of a property using a **Price Prediction Model**. By entering certain criteria like property type, size, and location, visitors can get an estimated market price for the property, helping them make informed decisions before taking the next steps in the process.

- **Display Property List**

  Upon fetching, the properties are displayed to the visitor in an organized and readable format that highlights essential details such as property type, price, location, and unique selling features. Additionally, visual elements like highquality images and interactive maps may be incorporated to provide a more immersive browsing experience, allowing visitors to virtually explore each property before making inquiries or scheduling viewings.

- **Admin Perspective**

  For administrators, the access to the system is designed with a higher level of security and functionality, as they are responsible for managing the properties and ensuring the accuracy of the information provided to visitors. The workflow for admins is structured and consists of several key components:

Login Process

- **Verify Credentials**

  To access the admin panel, administrators must first log in by entering their credentials, specifically a username and password. The system rigorously checks these credentials against its database to ascertain user authenticity.

- **Return Verification Status**

  Upon verification, the system responds with a status message.

  - **Success:** If the credentials are valid, access is granted, allowing the admin to proceed with various management functions.
  - **Failure:** Conversely, if the credentials are incorrect, access is denied, and the system prompts the admin to retry, thereby maintaining a secure environment.

Property Management

Once logged in, admins can perform a series of critical CRUD (Create, Read, Update, Delete) operations that are vital for effective property management:

- **Add Property**
  - **Insert Property**: Admins can input new property details into the system, ensuring that all relevant information, such as property descriptions, pricing, and images, are captured.

- **Confirm Insertion**: The system then validates the entered data for accuracy and completeness before saving it to the database.
- **Property Added**: Upon successful addition, a success notification is displayed, confirming that the property is now live on the platform.

- **Edit Property**
  - **Update Property**: Admins have the capability to modify existing property details, allowing them to keep listings accurate and up to-date as conditions change.
  - **Confirm Update**: The system will validate the changes made to ensure data integrity.
  - **Property Edited**: Following a successful update, a notification is shown to confirm that changes have been successfully implemented.

- **Delete Property**
  - **Remove Property**: Admins can select properties that need to be removed from the listings.
  - **Confirm Deletion**: The system will prompt for verification to prevent accidental deletions.
  - **Property Deleted**: After successful deletion, a notification confirms that the property has been removed from the database.

- **View Properties**
  - This function mirrors the visitor flow, allowing admins to fetch and review the property list, ensuring they stay informed about all current listings available to visitors.

- **Key Features**

  The system is equipped with several key features designed to enhance functionality and security:

  - **Role-Based Access**: The system clearly delineates user roles, allowing visitors only to view properties while giving admins the necessary tools to manage these listings effectively.
  - **Authentication**: A secure login process ensures that only authorized admins can access sensitive data and modify property information, thereby safeguarding the integrity of the system.
  - **CRUD Operations**: The system provides clear and straightforward steps for adding, editing, deleting, and viewing properties, making management tasks efficient and user-friendly.
  - **Confirmation Steps**: Each action, whether it be adding, editing, or deleting a property, includes validation mechanisms to maintain data accuracy and integrity, preventing accidental data loss or errors.

- **System Flow**

  The system flow is designed to cater to different user needs, with a clear distinction between visitors and admins:

  - **Visitor**: Visitors experience simple read-only access, allowing them to browse properties freely without the ability to make changes.
  - **Admin**: Admins engage in a structured workflow that begins with login, followed by the performance of various actions such as adding, editing, or deleting properties, and concludes with a logout process (implied for security reasons).

This diagram efficiently separates user roles and highlights the backend processes, such as database interactions, for each action, ensuring a clear understanding of the system's operation and user interactions.

4.5.3 Class Diagram

The attributes and methods of each class within the system can be further defined based on specific requirements and implementation details, leading to a comprehensive class diagram that visually represents the relationships and functionalities of the various components involved in property management. This diagram will serve as a foundational guide for developers and stakeholders alike, ensuring clarity and consistency in the development process:

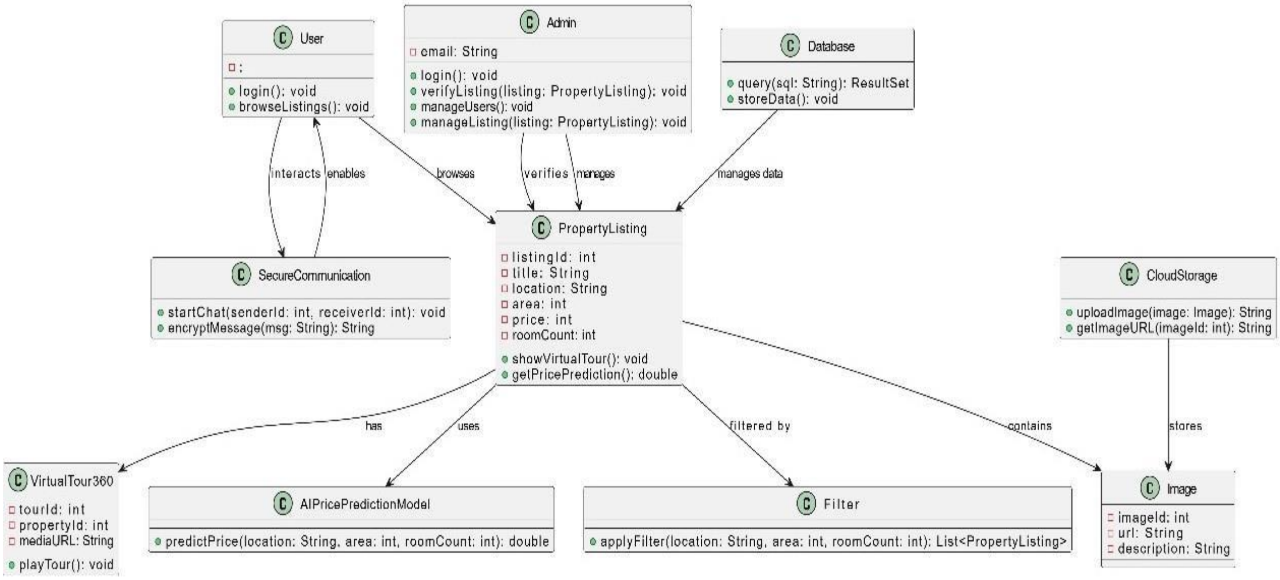

Figure 4 Class Diagram of Real Estate System

Chapter 5
Implementation

5 Implementation

In the implementation phase of **Home2U**, the website is developed using React.js for the frontend, offering a dynamic and responsive user interface. The backend is powered by Sails.js, a robust MVC framework built on Node.js, which facilitates scalable and maintainable server-side logic. Home2U is a web-based platform only, meaning it does not require any installations or specific configurations. Users can access it seamlessly through any modern web browser on desktop, tablet, or smartphone. The primary goal of this platform is to provide users with up-to-date property listings, allowing them to explore and take advantage of the latest opportunities in the real estate market. The screenshots below are actual images taken from the live site, illustrating its intuitive layout and key features. Home2U is designed as a web-based platform only, which means it does not require users to go through any installation processes, software downloads, or specific configurations to begin using it. Instead, users can conveniently access the full functionality of the platform directly through any modern web browser, whether they are on a desktop computer, tablet, or smartphone. This cross-device compatibility ensures that the application remains accessible and usable from virtually any location with an internet connection. The primary objective of the Home2U platform is to serve as a comprehensive digital solution that provides users with real-time, up-to-date property listings**,** empowering them to explore the latest opportunities and developments in the rapidly changing real estate market. The screenshots provided below are actual images captured from the live website, showcasing its user-friendly layout, visually clear design, and key features that help users engage effectively with the platform.

5.1 Admin login Algorithm

Home2U includes a secure admin login system that allows only authorized users to access the backend, ensuring that sensitive data and functionalities are protected from unauthorized access. Admins can add, update, or remove property listings with ease, ensuring the website content stays accurate and up to date, which is crucial in the fast-paced world of real estate where timely information can make a significant difference.

5.2 External APIs

5.2.1 Google OAuth

To enhance functionality and security, Home2U leverages several external APIs. For secure admin login authentication, it utilizes **Google OAuth** (Google Developers, 2023), which provides a robust mechanism for user identity verification. This component internally uses

- https://accounts.google.com/o/oauth2/auth – for initiating the OAuth 2.0 authorization request (user consent screen)
- https://accounts.google.com/o/oauth2/token – for exchanging the authorization code for access and ID tokens (if using the code flow)

5.2.2 EmailJS

Furthermore, **EmailJS** (EmailJS, 2023) is utilized to handle contact form messages via POST requests, facilitating efficient communication between users and the platform. This is achieved through the endpoint: https://api.emailjs.com/api/v1.0/email/send.

5.2.3 Amazon S3

Amazon S3 is deployed for image storage and retrieval, ensuring that property images can be accessed quickly and reliably. The API endpoints for image handling include both GET and POST requests: https://dreamdwellnessrealestate.s3.amazonaws.com/properties/

5.3 User Interface

The Home2U website features a clean and intuitive user interface (UI) that is thoughtfully designed to significantly enhance the overall user experience by providing a smooth, simple, and user-friendly interaction throughout the platform. With its easy navigation system, users can quickly move between different sections of the website without confusion, making the browsing experience seamless and efficient. The site incorporates advanced filters that enable users to refine their search results based on specific criteria such as location, price range, property type, and more, helping them find the most relevant listings according to their unique preferences. Additionally, the property listings are presented in an organized and visually appealing manner, allowing users to quickly assess key information about each property at a glance. The platform also includes a client-side contact form, giving visitors a straightforward way to get in touch without navigating away from the current page, further emphasizing user convenience and accessibility.

Built entirely with the user in mind, the platform leverages a sophisticated Random Forest model that has been trained to predict property price ranges with a commendable accuracy of 71%, making it a valuable tool for users to estimate fair market values based on different property features and characteristics. Although the client reviews are currently hardcoded, they serve as an effective form of initial social proof, offering visitors insights into positive user experiences and helping to build trust in the platform. Users can easily browse or manage listings through visually engaging and responsive layouts that are designed to adapt well across different devices and screen sizes, ensuring a consistent experience for all users. Furthermore, the footer section of the website is thoughtfully designed to include all essential contact information, links to social media profiles, and previews of the latest property listings, ensuring that users have access to important resources and recent updates no matter where they are on the site.

5.3.1 Welcome/navigation

The Welcome and Header screen of the Home2U website offers a clean and visually appealing entry point, setting the tone for the overall user experience right from the start. It features a well-organized top navigation bar that is designed to provide users with quick and easy access to various important sections of the site, such as property listings, about us, and contact information. In addition to the navigation links, the header includes essential contact info that allows users to reach out quickly for inquiries or assistance, making communication with the platform simple and efficient. Moreover, an admin login icon is also prominently displayed in the header, providing authorized personnel with secure access to the administrative panel for managing content and listings.

The main banner on the Welcome screen is designed to immediately catch the attention of visitors by prominently highlighting featured properties, especially those that are available "For Sale". This clear and bold label draws user attention to key listings and helps guide them toward relevant real estate opportunities. The banner is further enhanced by a catchy tagline that effectively communicates the platform's core purpose helping users find homes in Pakistan creating a sense of trust and relevance for the local market. Additionally, the banner includes a starting price point that is strategically placed to attract the interest of potential buyers, offering a glimpse of affordability and value. Together, these elements create an inviting and informative first impression that encourages users to explore the rest of the platform.

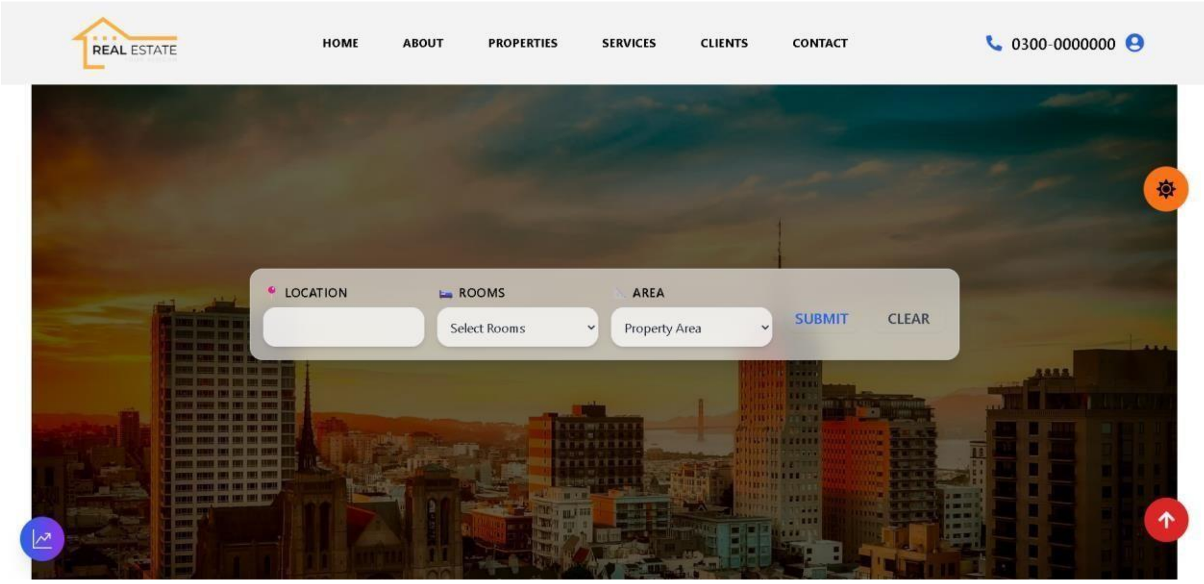

Figure 5 Welcome Screen

5.3.2 Filter

To enhance the overall user experience, the Home2U website includes a simple yet effective filter bar that is thoughtfully designed to make the property search process faster, easier, and more efficient for all types of users. This feature enables users to easily search for properties by entering specific criteria such as the location, the number of rooms, and the property area, which are common and important filters used by people looking for a new home or investment. The intuitive layout and accessibility of the filter bar ensure that users of all technical backgrounds can take advantage of this functionality without confusion or unnecessary steps.

The design of the filter bar is user-friendly and provides clear options, allowing users to apply the selected filters by clicking the Submit button, which instantly displays results matching their preferences. In addition to this, a convenient Clear option is also provided so users can quickly remove all filters and start a new search from scratch, without having to reload the page or manually undo previous selections. This thoughtful combination of features makes it easy to facilitate a quick and efficient way for users to discover suitable properties that match their specific needs and preferences, ultimately improving their experience on the platform and making property exploration more streamlined and enjoyable.

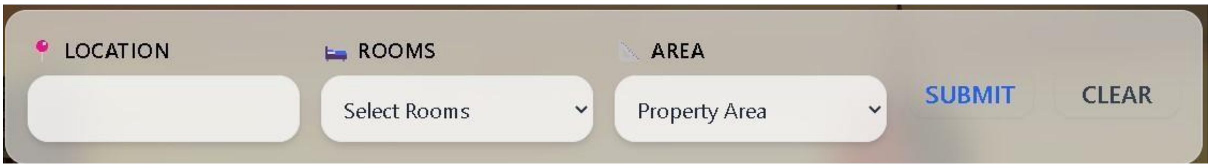

Figure 6 Search Filters

5.3.3 Model

A Random Forest Regressor was meticulously trained using various real estate features such as city, property type, area, and number of rooms after encoding categorical data. The predictions for property prices were converted into user-friendly bins (e.g., "Around 10 Lakh") utilizing defined ranges, making it easier for users to understand potential price points. The model's achievement of 71% accuracy in predicting the correct price range underscores its reliability and effectiveness in assisting buyers. Once trained, the model's predictions for property prices were converted into user-friendly bins to enhance usability and understanding for general users. These bins included descriptive labels such as "Around 10 Lakh", created using defined price ranges to represent various segments of the real estate market. This approach helped make the pricing data more accessible and intuitive for individuals who may not be familiar with technical pricing figures. The model's achievement of 71% accuracy in predicting the correct price range is a testament to its reliability and effectiveness, showing that it can serve as a valuable tool for assisting buyers in making informed decisions when exploring property options on the platform.

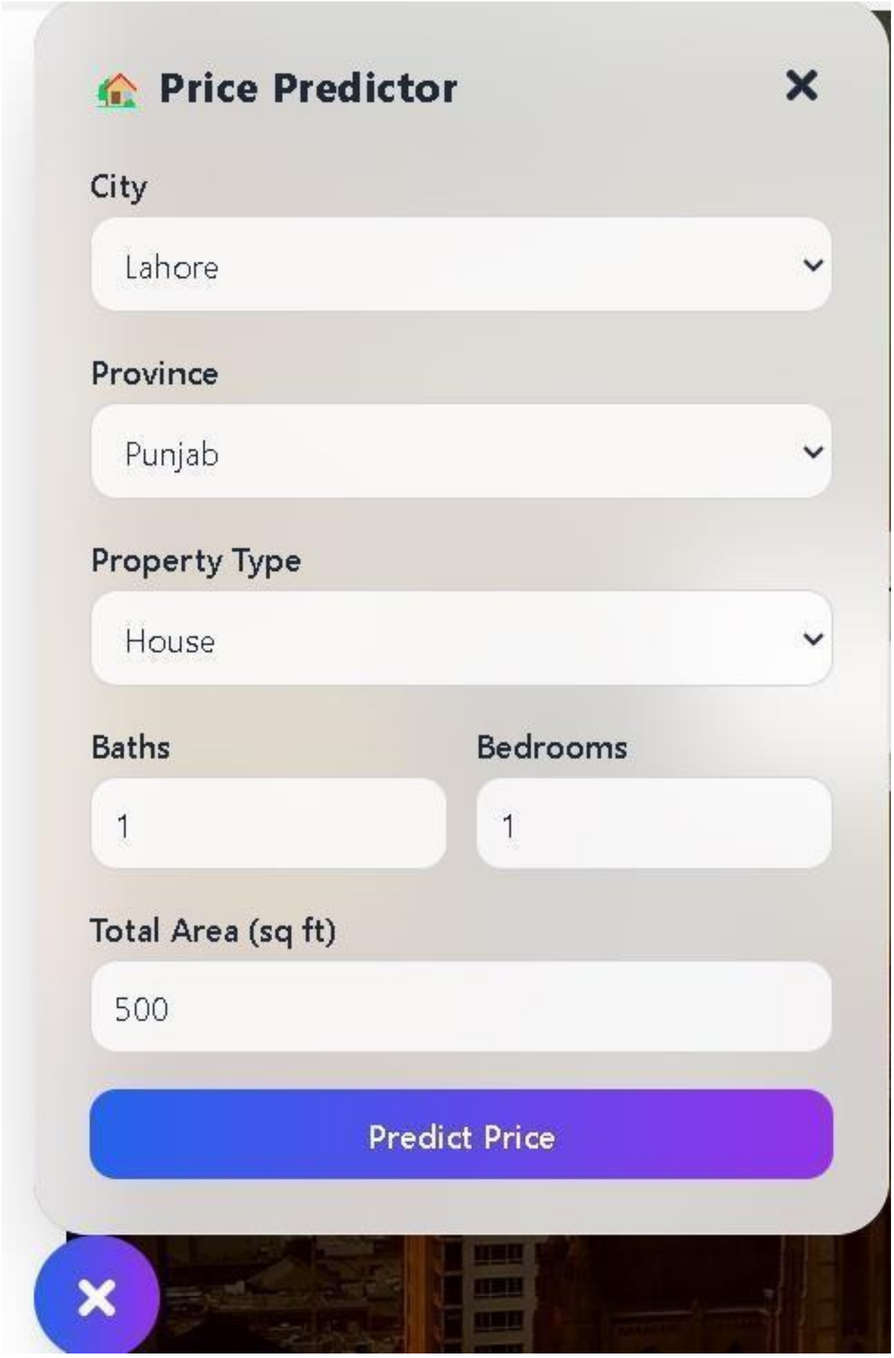

Figure 7 Price Prediction

5.3.4 Properties

The interface showcases a dedicated section titled "Properties," which serves as a central hub for users interested in browsing or managing real estate listings on the platform. Accompanied by the subheading "Explore the Latest," this section actively encourages user interaction by prompting visitors to log in to add or edit properties, thereby maintaining a secure and user specific environment. This login requirement ensures that only authorized users can modify listings, which helps preserve the integrity and accuracy of the property information displayed on the website. The layout and organization of this section are intentionally designed to make navigation simple and intuitive, ensuring that users can easily find what they are looking for.

Within this section, three exemplary properties are highlighted to demonstrate the kinds of listings available. Each of these properties is presented with distinctive titles that reflect their key features, along with eye-catching images that offer visual appeal and instantly attract attention. These listings also include additional details such as property IDs for easy identification, as well as comprehensive descriptions that give users a better understanding of each property's unique attributes and selling points. The layout is crafted for interactivity, integrating icons for actions like edit, share, and like buttons, all of which work to enhance user engagement by providing immediate, accessible tools for interaction. The overall design follows a user-centric approach, prioritizing ease of use and visual clarity. It incorporates visual storytelling elements that not only inform users but also captivate visitors and inspire further exploration of the platform's offerings.

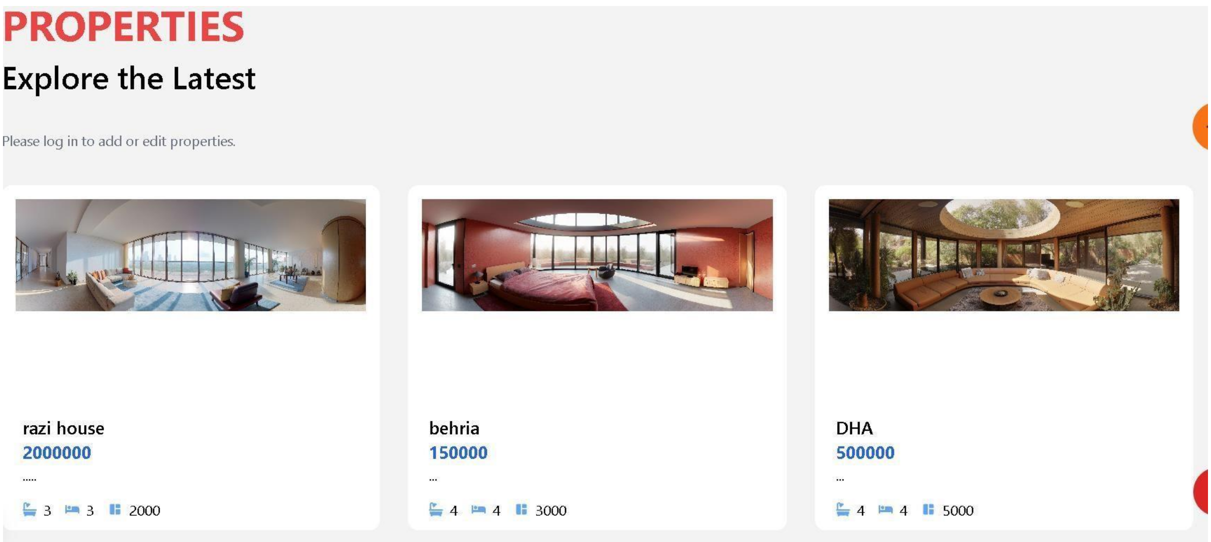

Figure 8 List of Properties

5.3.5 Reviews

The "Our Clients" section of the website currently features hardcoded testimonials, where each review includes the client's name, photograph, feedback, and star rating, all manually entered into the frontend code. This approach, while straightforward in development, ensures consistent design and presentation. However, it does present limitations regarding scalability and flexibility, as any addition or modification of reviews necessitates code changes. For future iterations, integrating a backend or database to dynamically fetch testimonials could significantly improve maintainability and enhance user experience. As the number of client testimonials grows or changes over time, this hardcoded setup means that any new addition, removal, or modification of a review would require manual updates to the source code itself. This dependency on developer intervention can slow down content updates, increase the chance of errors, and make the system less adaptable to frequent changes. To overcome these limitations and prepare for future needs, it would be beneficial to consider integrating a backend system or database solution. By dynamically fetching testimonials from a centralized data source, the platform could significantly improve long-term maintainability, streamline the content management process, and offer a more seamless and engaging user experience for visitors interacting with the testimonials section.

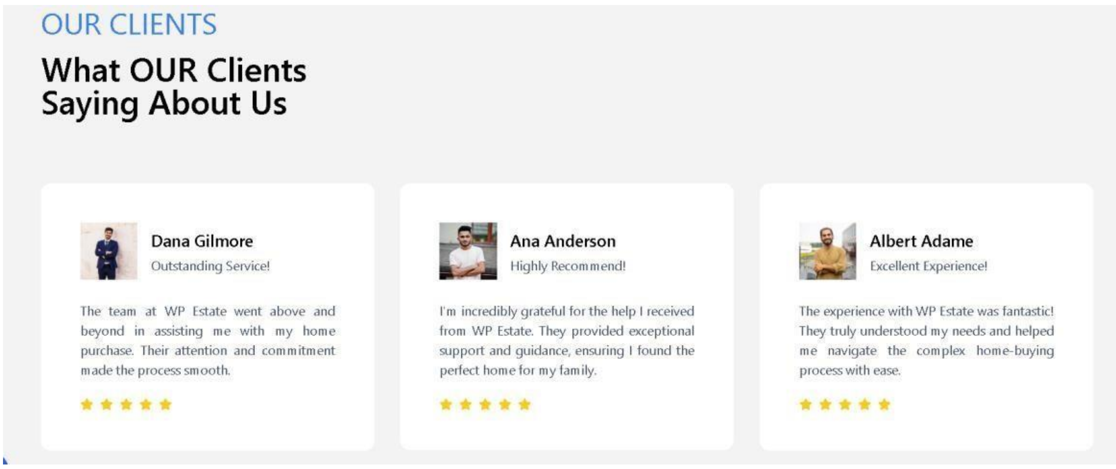

Figure 9 Client Reviews

5.3.6 Contact/Complain

The contact form is implemented purely on the frontend, facilitating users to send messages without the need for a server. It includes fields for name, subject, email, phone number, and message, allowing for comprehensive communication. This setup likely utilizes a third-party service, such as EmailJS, for sending emails directly from the client side, ensuring a lightweight and serverless communication process that streamlines user interactions.

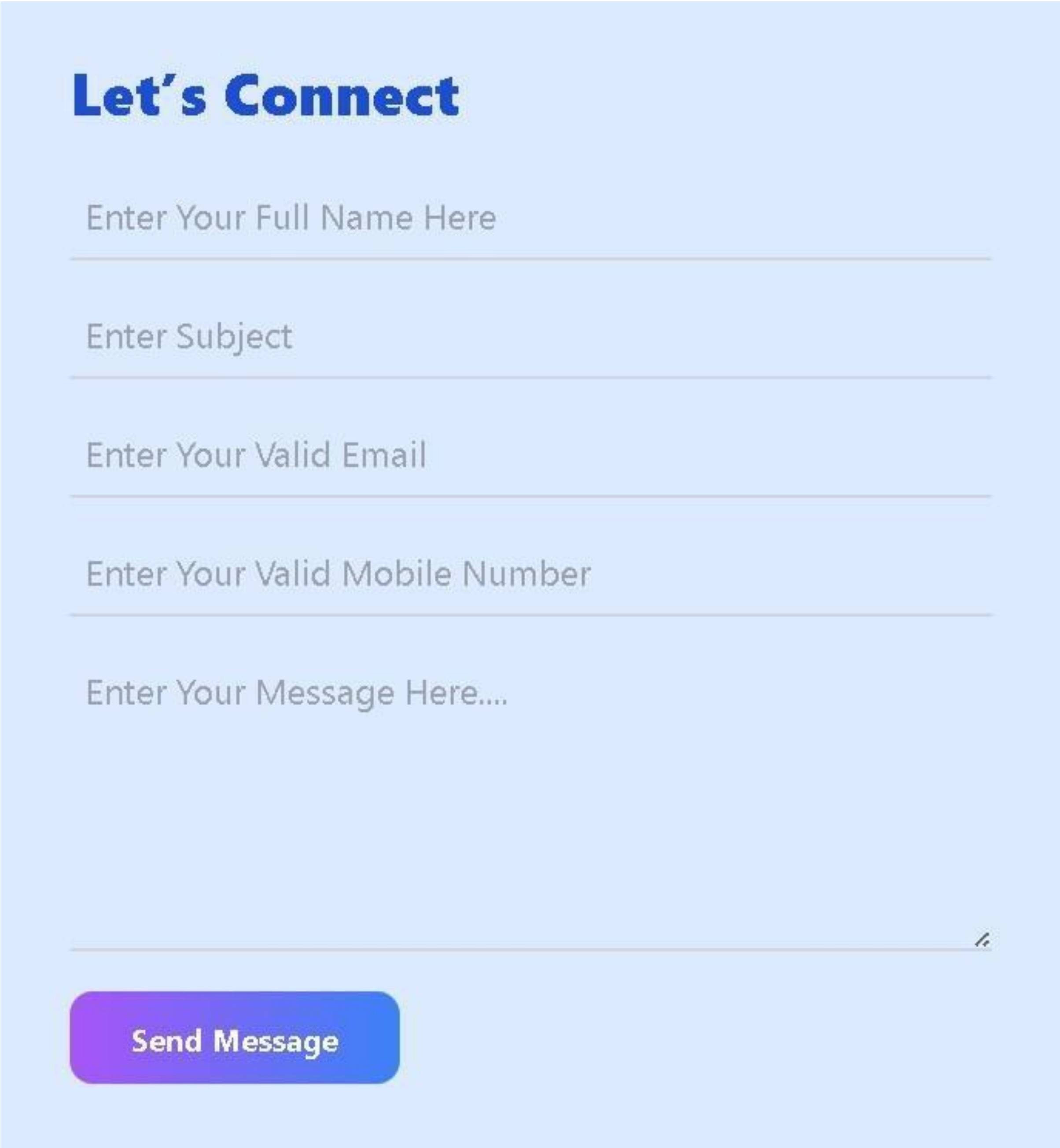

Figure 10 Contact and Complaint

5.3.7 Footer

The footer contains three main components essential for user engagement:

- **Social Media Links**

  These links allow users to connect with Home2U via popular platforms such as Facebook, Instagram, Twitter, and YouTube, fostering community and brand presence.

- **Contact Information**

  This section displays the business address, phone numbers, and email, ensuring that users can easily reach out for inquiries or support.

- **Latest Properties**

  This feature provides previews of recent property listings, complete with images and prices, encouraging users to explore new opportunities as they arise.

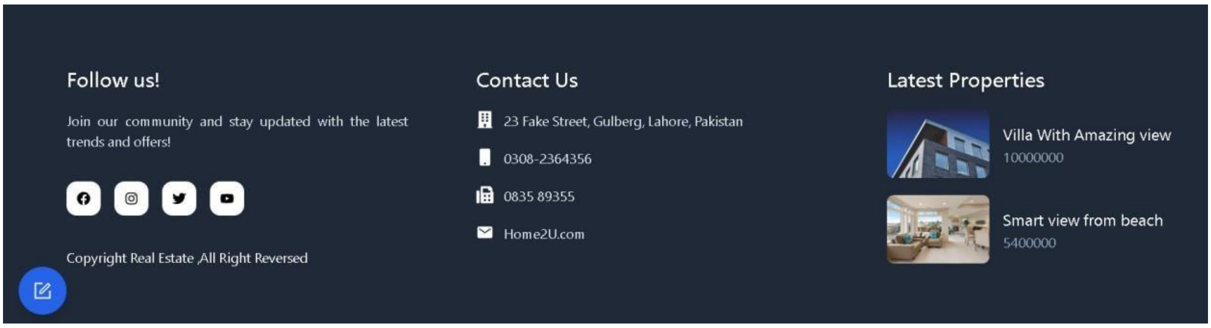

Figure 11 Footer

5.3.8 Property details

This is a **Property Details Page** from a real estate website. Here's a breakdown of what it shows:

- **Property Name:** DHA
- **Location:** Islamabad
- **Price:** $500,000
- **Owner:** Noman
- **Features**
  - 4 Bedrooms
  - 4 Bathrooms
  - 5000 sq ft area
- A large image showing the interior view of the property.

- A **"Back to Listings"** button to return to the full property listings page.
- A top navigation bar with links like **Home**, **About**, **Properties**, **Services**, **Clients**, and **Contact**.
- Footer section includes social media follow options, contact details, and a latest properties preview.

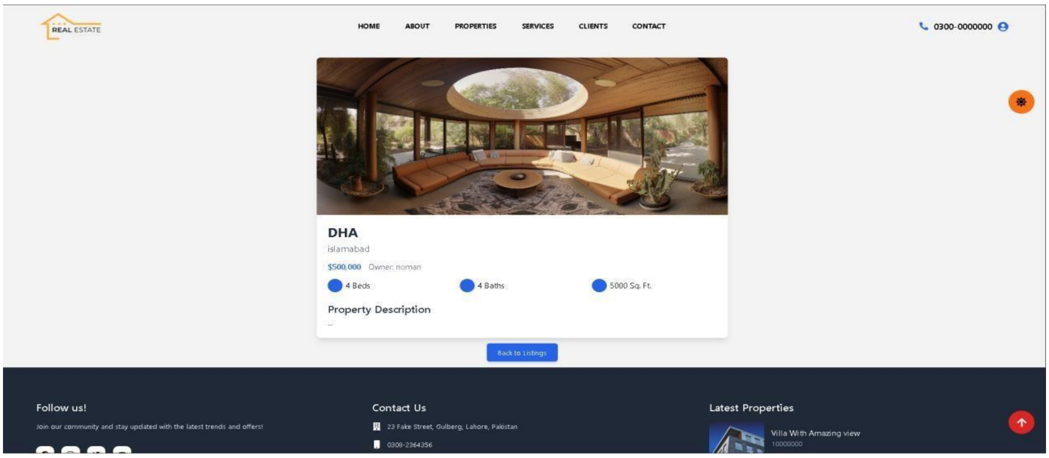

Figure 12 Property Detail

Chapter 6
Testing and Evaluation

6 Testing and Evaluation

6.1 Unit Testing

Unit testing was done to validate real-world interaction and user flow.

Table 10 Unit Testing

Test Case ID	Feature	Test Scenario	Expected Outcome	Actual Outcome
UT-001	Image Upload	Store property image to Amazon S3.	Image is uploaded and link is saved.	Image successfully uploaded.
UT-002	Data Storage	Save property data (price, address, etc.).	Data saved correctly in the database.	Data saved successfully.
UT-003	Price Prediction	Predict price using model based on inputs.	Predicted price is returned accurately.	Model returned correct price bin.
UT-004	Admin Login	Login with Google OAuth.	Admin is authenticated.	Login successful.
UT-005	EmailJS Integration	Send contact form submission via EmailJS.	Email is sent successfully.	Email sent and received.

6.2 System Testing

Complete end-to-end testing of Home2U platform.

Table 11 System Testing

Test Case ID	System Components	Test Case Description	Expected Result	Actual Result

ST-01	Full Workflow	Visit site, browse, contact, admin login, add.	All functionalities work without error.	System works perfectly.
ST-02	High Load Simulation	Load test with 100+ concurrent users.	System remains stable.	No crashes, good performance.
ST-03	Failure Recovery	Simulate email or DB failure.	System handles errors gracefully.	Resilient and selfrecovering.

6.3 Manual Testing

Manual testing was done to validate real-world interaction and user flow.

Table 12 Manual Testing

Test Case ID	Feature	Test Scenario	Expected Outcome	Actual Outcome
MT01	Homepage Navigation	Browse homepage, view banner, listings.	Homepage loads with all sections.	Homepage loads perfectly.
MT02	Admin Login	Login as admin.	Successful login with dashboard access.	Login and access granted.
MT03	Add Property	Admin adds property with image + details.	Property appears in listings.	Property added successfully.
MT04	Contact Form	Fill and submit form using EmailJS.	Email is sent and acknowledged.	Working as expected.

MT05	Image Rendering	Check image load via S3 URL.	Image loads correctly in UI.	Image displays properly.

6.4 Functional Testing

Ensures complete system behavior aligns with business requirements.

Table 13 Functional Testing

Test Case ID	Functionality	Test Case Description	Expected Result	Actual Result
FT-001	Admin Registration	Google OAuth for login.	Login authenticated with Google.	Login successful.
FT-002	Property Management	Add/update/delete properties.	CRUD operations work.	Fully functional.
FT-003	Image Upload to S3	Upload and retrieve images.	Image URL is generated and retrievable.	Image accessible.
FT-004	Price Prediction	Predict price based on property details.	Price returned in readable format.	in Returned correct range.

6.5 Integration Testing

Tests integration of backend, frontend, and third-party services.

Table 14 Integration Testing

Test Case ID	Components Involved	Test Case Description	Expected Result	Actual Result
IT-001	Form + EmailJS	Contact form triggers email on submit.	Email sent and confirmation shown.	Works correctly.

IT-002	Upload + S3 + Database	Upload image, save URL in DB.	Image stored and URL saved.	Working as expected.
IT-003	Prediction Model + UI	Submit data to model, show price range.	Price returned to UI properly.	Prediction shown successfully.

Chapter 7
Conclusion & Future Work

7 Conclusion and Future Work

7.1 Conclusion

The Home2U real estate platform has undergone comprehensive and thorough testing across multiple levels to ensure its stability, accuracy, and performance. These levels include unit testing, manual testing, functional testing, integration testing, system-level testing, and automated testing. Each aspect and feature of the system, ranging from critical components like image handling through Amazon S3 to the admin login functionality facilitated via Google OAuth and the integration of the contact form using EmailJS, has been rigorously validated and tested to confirm their reliability, responsiveness, and overall efficiency. Particular attention was given to the property pricing prediction model, which was assessed for its precision, accuracy, and relevance to real-world market data, and it consistently produced outputs that were meaningful, useful, and reliable for end users.

The testing outcomes and results clearly demonstrate that the application successfully fulfills all of its defined functional requirements. It delivers a smooth and intuitive user experience across devices and platforms, ensuring that users can navigate and interact with the system effortlessly. Additionally, it shows dependable performance under various real-world usage scenarios, validating its practical value and usability. The system is built with robust and reliable error handling mechanisms, and its architecture is designed to scale effectively with growing demand. Combined with a modern and clean user interface design, these features collectively ensure a pleasant and effective user experience. As a result of these qualities, the Home2U platform is now fully production-ready and well-positioned to support the evolving needs of users who wish to list, browse, or explore property options. It is prepared to confidently meet the demands of the dynamic and rapidly growing real estate market in Pakistan, establishing itself as a reliable and scalable digital solution.

7.2 Future Work

Looking ahead, several enhancements are planned to further enrich the Home2U platform and provide users with an even more comprehensive and engaging experience

7.2.1 Dynamic User Registration and Login

What

Extend the platform to allow buyers and sellers to register, log in, and manage their own property listings, fostering a more personalized user experience.

How

- Add registration and login pages using Firebase Auth or the existing Google OAuth setup, ensuring secure and user-friendly authentication.
- Implement role-based access to clearly separate admin functionalities from regular user capabilities, enhancing security and usability.
- Store user profiles and session data in your existing database to streamline user interactions and maintain continuity across sessions.

Where

This feature can be integrated into the frontend (React) with protected routes for users and in the backend to manage user roles, ensuring a seamless experience for all users.

7.2.2 Interactive Chat with Agents

What

Introduce a real-time chat feature that allows users to communicate directly with property agents to ask questions or arrange site visits, creating a more interactive platform.

How

- Use a real-time communication library such as Socket.io or Firebase Realtime Database to facilitate immediate messaging capabilities.
- Create a user-friendly chat interface on property detail pages to enhance communication efficiency.
- Link each chat session to the logged-in user's ID and agent profile, ensuring personalized interactions.

Where

This feature will require integration of both frontend (React components) for user interaction and backend (Node.js or similar) for managing real-time messaging and chat logs.

7.2.3 AI-Based Property Recommendations

What

Develop an intelligent recommendation system that suggests properties to users based on their browsing behavior, search history, or past interactions, enhancing user engagement and satisfaction.

How

- Collect behavioral data such as clicked properties, filter usage, and time spent on listings to tailor recommendations effectively.
- Employ a recommendation algorithm (e.g., collaborative filtering or content-based filtering) to analyze user preferences and provide relevant suggestions. (Scikit-learn, 2023)
- Display suggested listings prominently on the homepage or user dashboard to encourage exploration of properties that align with user interests.

Where

Backend services will process the data collected, while frontend components will dynamically present "Recommended For You" sections based on user activity.

7.2.4 Analytics Dashboard for Admins

What

Create an advanced analytics dashboard specifically for admins, providing visual insights into property trends, user activity, and overall system performance, empowering data-driven decision-making.

How

- Utilize charting libraries (like Chart.js or Recharts) in the admin panel to present data visually, making it easier to interpret key metrics.
- Pull data from your database, including views, clicks, contact submissions, and other relevant statistics, to inform admins about user engagement and system health.
- Display metrics such as top-viewed properties, prediction model accuracy, and contact form usage for comprehensive oversight of platform dynamics.

Where

This dashboard will be integrated into the admin portal, ensuring access is restricted to authenticated admins, thereby maintaining data security

By implementing these enhancements, Home2U will not only elevate user experience but also streamline administrative operations, ultimately positioning itself as a leader in the real estate digital landscape.

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